

Nmap : Network Scanner

Introduction:

- All machines connected to a LAN or connected to the Internet via a modem run many services that listen at certain ports.
- A service is a program that waits inside a loop for a request message from a client, and acts on the request.
- By port scanning, one discovers which ports are available (i.e., being listened to by a service). Essentially, a port scan consists of sending a message to each port, one at a time and examining the response received.
- If the port is in use, it can then be probed further for weakness. Port Scanning is one of the most popular among the reconnaissance techniques attackers use.

The port numbers are divided into three ranges:

- 1- Well Known Ports (from 0 through 1023)
- 2- Registered Ports (from 1024 through 49151)
- 3- Dynamic and/or Private Ports (from 49152 through 65535)

Sockets:

- A socket is an abstraction, similar to a file descriptor, constructed by `socket()`.
- A socket so constructed is bound to an IP address and port number via the `bind()` call.
- A server program then waits for a connection via the `listen()`, and `accept()`s a connection

Open Port: A service process is listening at the port. A port is opened by the OS at the request of a specific process. The OS receives packets arriving at this port and gives the messages to the service process. If the OS receives a SYN at an open port, this is the first packet of the three way handshake.

Closed Port: No process is listening at the port. If the OS receives a SYN at a closed port, an RST is sent.

Filtered Port: A packet filter is listening at the port.

Nmap: The nmap port scanner (www.nmap.org) is widely known. nmap is a utility for port scanning large networks, although it works fine for single hosts. Sometimes you need speed, other times you may need stealth. In some cases, bypassing firewalls may be required. Not to mention the fact that you may want to scan different protocols (UDP, TCP, ICMP, etc.). You just can't do all this with one scanning mode. And you don't want to have 10 different scanners around, all with different interfaces and capabilities. All these scanning technique you will found in nmap. Nmap ("Network Mapper") is an open source tool for network exploration and security auditing. It was designed to rapidly scan large networks, although it works fine against single hosts. Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics.

Uses of Nmap :

While Nmap is commonly used for security audits, many systems and network administrators find it useful for routine tasks such as network inventory, managing service upgrade schedules, and monitoring host or service uptime Nmap output : The output from Nmap is a list of scanned targets, with supplemental information on each depending on the options used. Key among that information is the "interesting ports table".

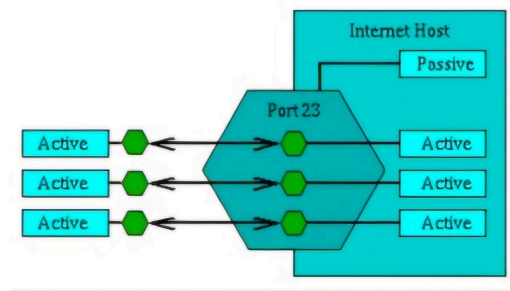


Figure 1

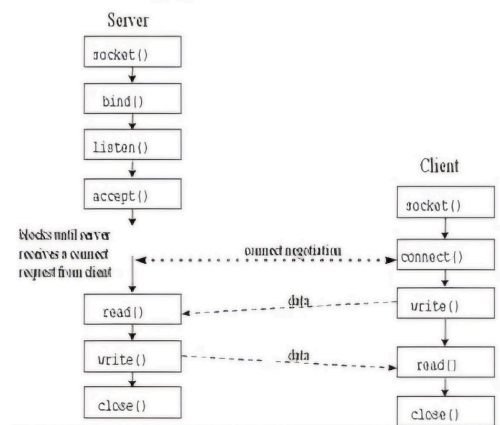


Figure 2 Socket calls for connection-oriented communication

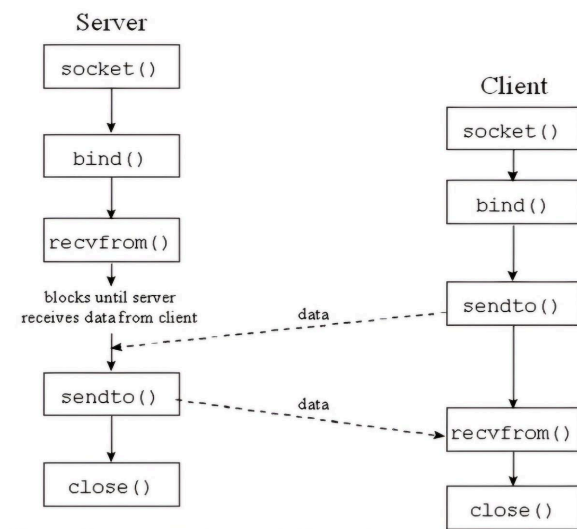
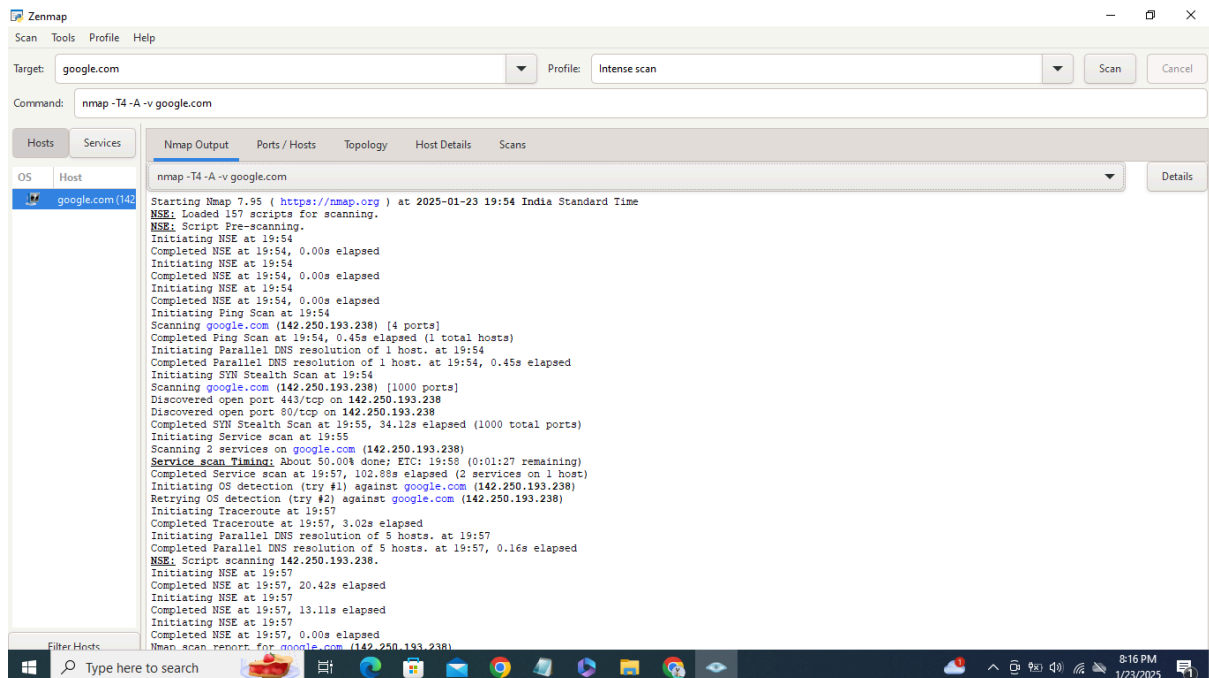
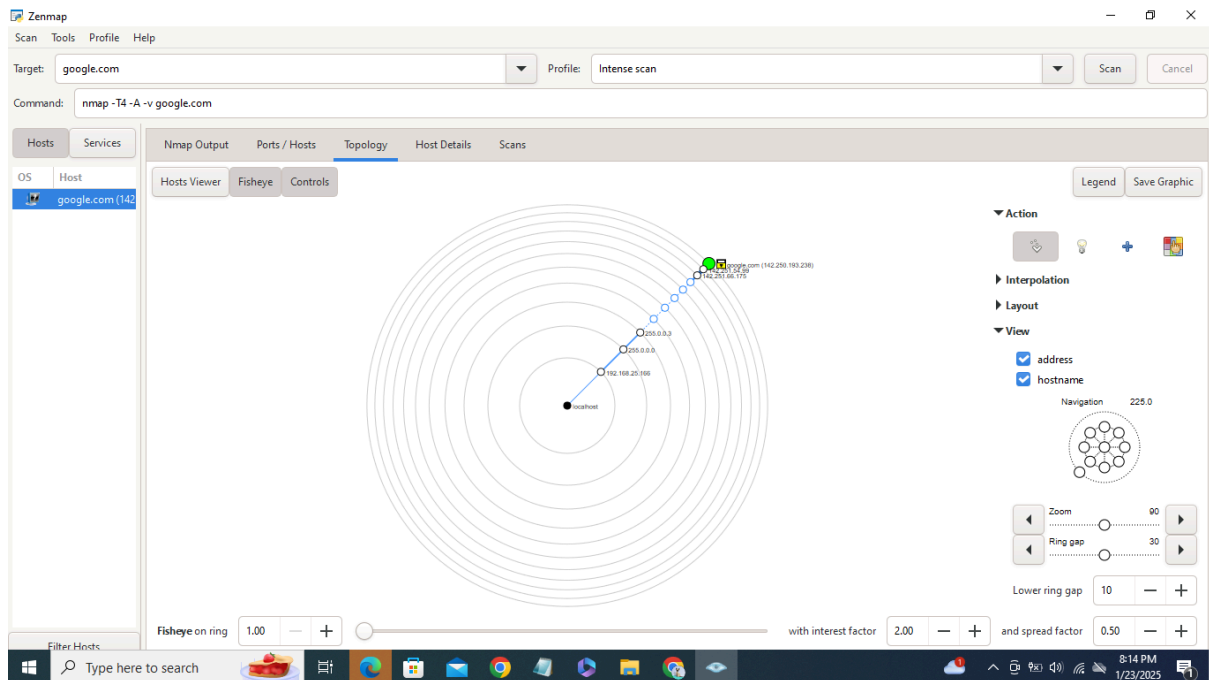
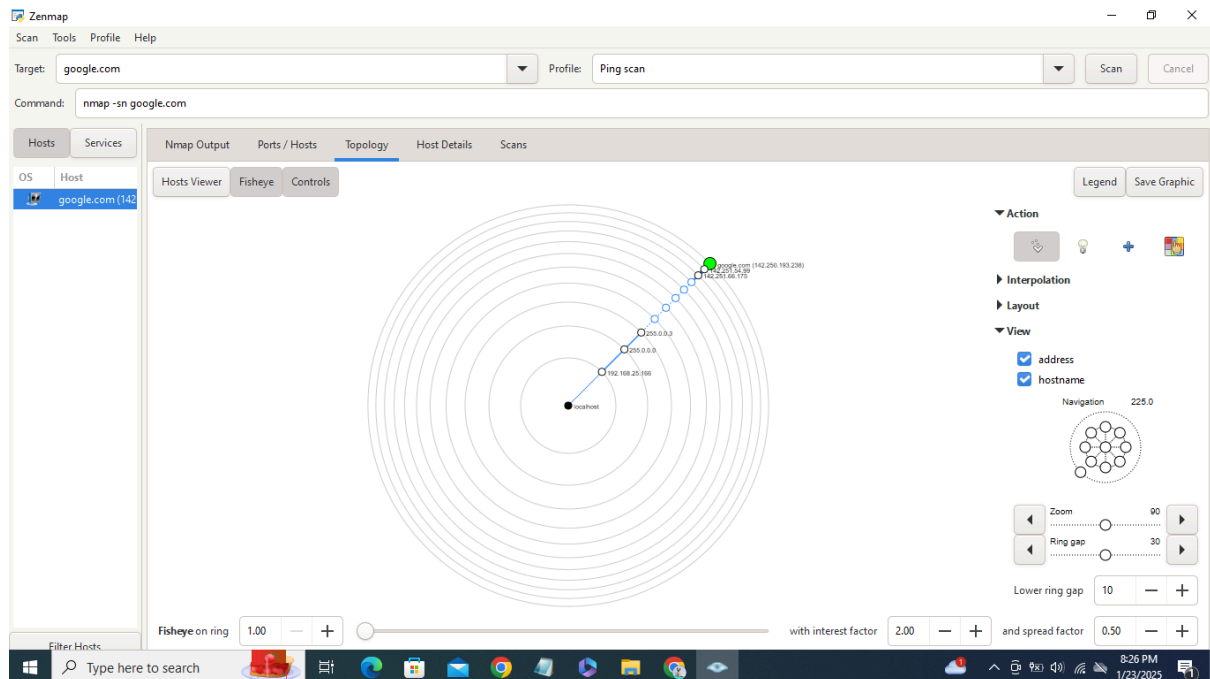
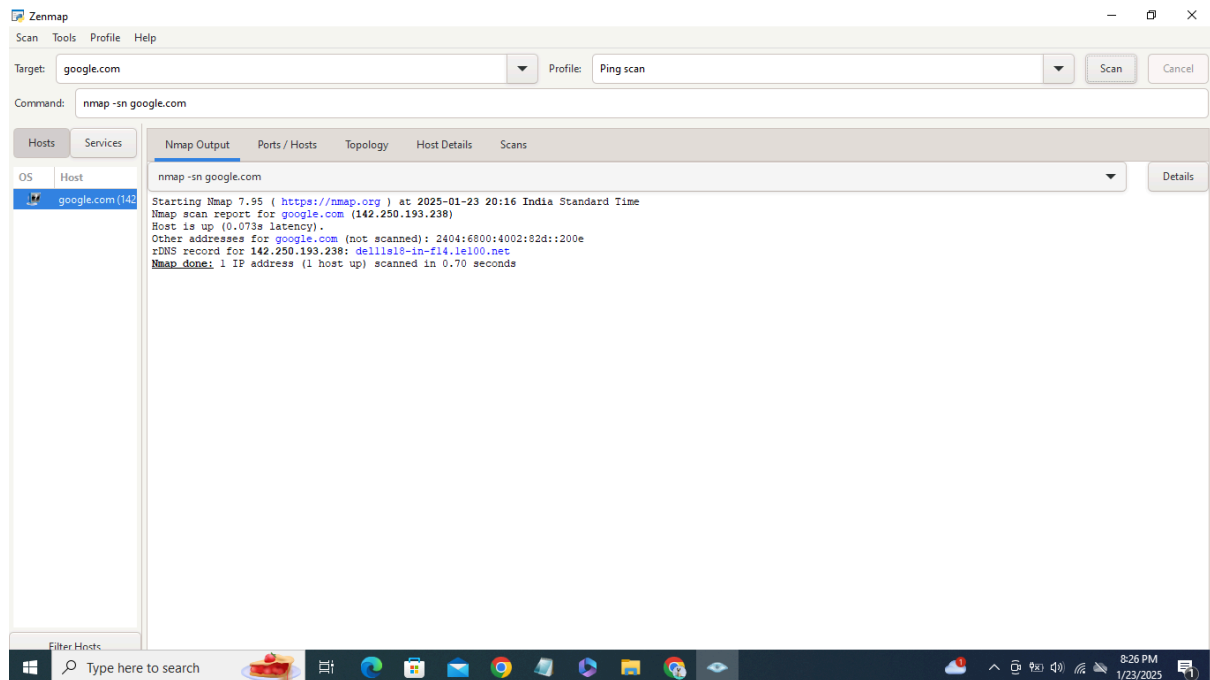


Figure 3 Socket calls for connectionless communication

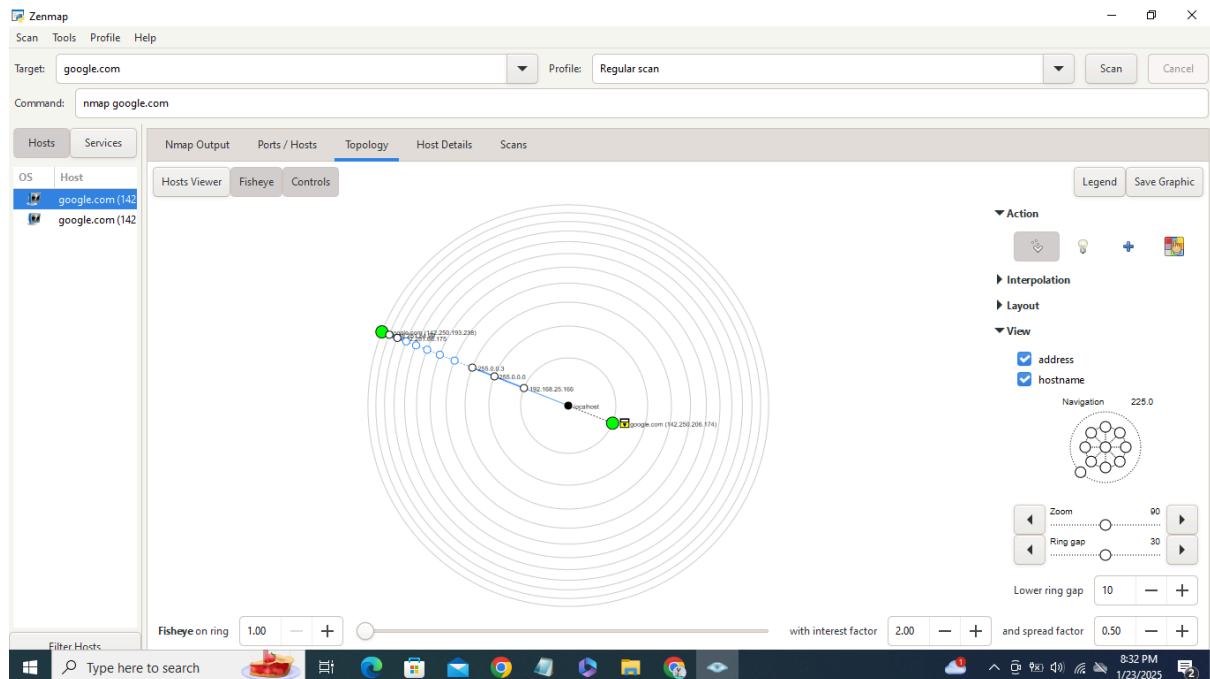
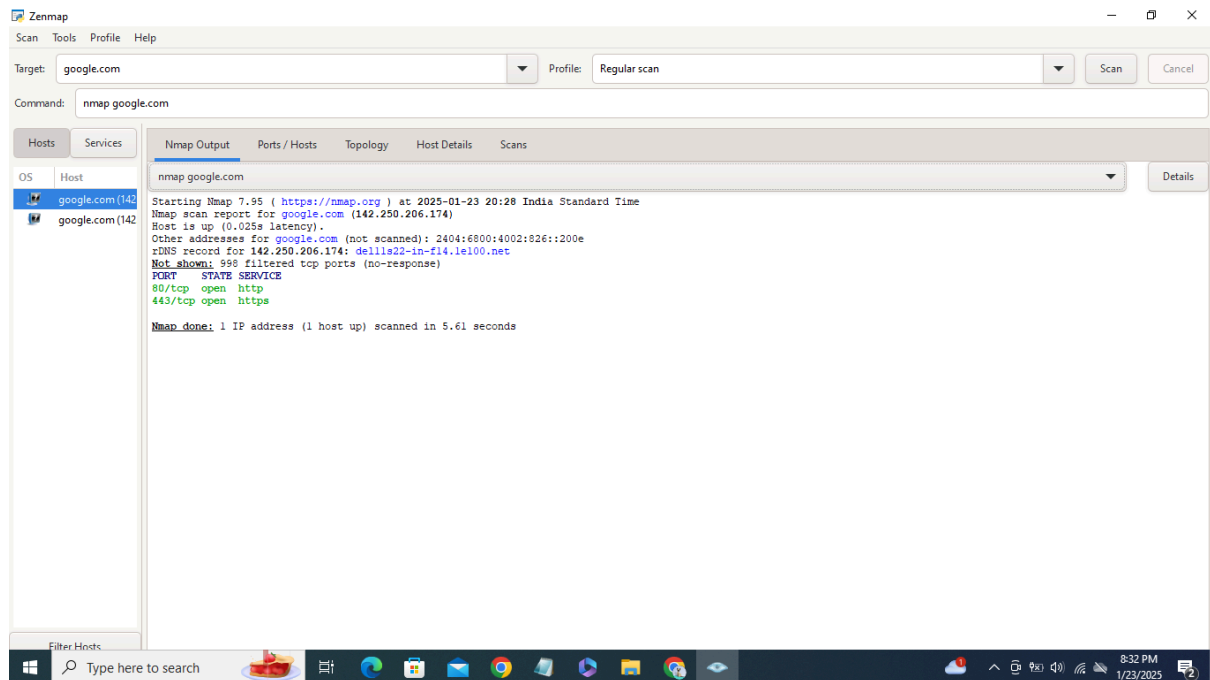
Intense Scan : Google.com



Google.com (Ping Scan)



Google.com (Regular Scan) :



Google.com (Quick Traceroute Scan) :

The screenshot shows the Zenmap application window. The 'Target' field is set to 'google.com' and the 'Profile' is 'Quick traceroute'. The command entered is 'nmap -sn --traceroute google.com'. The 'Nmap Output' tab is selected, displaying the following text:

```
nmap -sn --traceroute google.com

Starting Nmap 7.95 ( https://nmap.org ) at 2025-01-23 20:33 India Standard Time
Nmap scan report for google.com (142.250.193.238)
Host is up (0.024s latency).
Other addresses for google.com (not scanned): 2404:6800:4002:82b::200e
rDNS record for 142.250.193.238: dell1s18-in-fl4.1e100.net

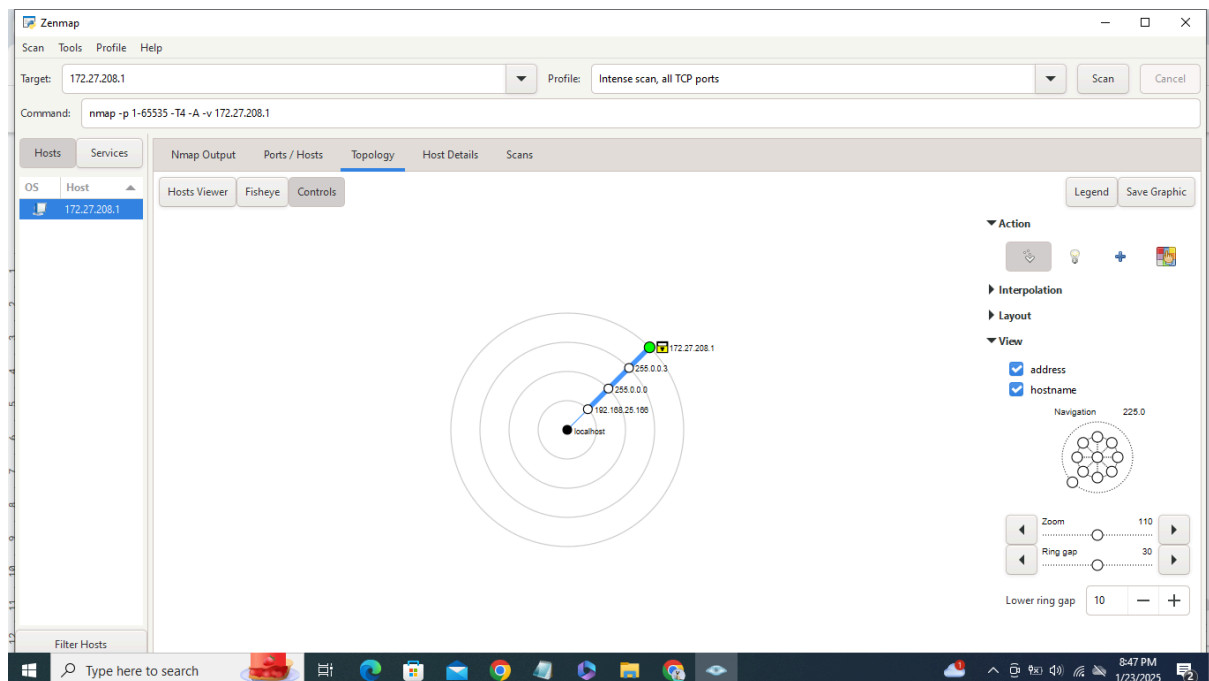
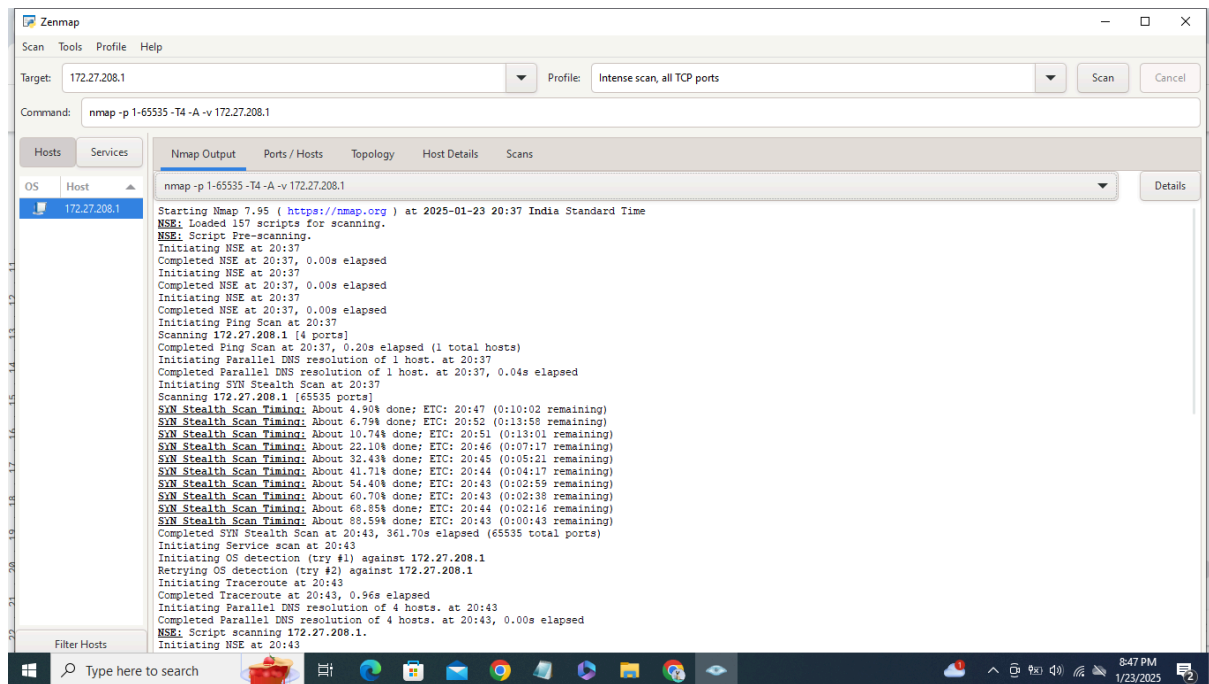
TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 4.00 ms 192.168.25.166
2 15.00 ms 255.0.0.0
3 16.00 ms 255.0.0.3
4 16.00 ms 255.0.0.3
5 16.00 ms dell1s18-in-fl4.1e100.net (142.250.193.238)

Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
```

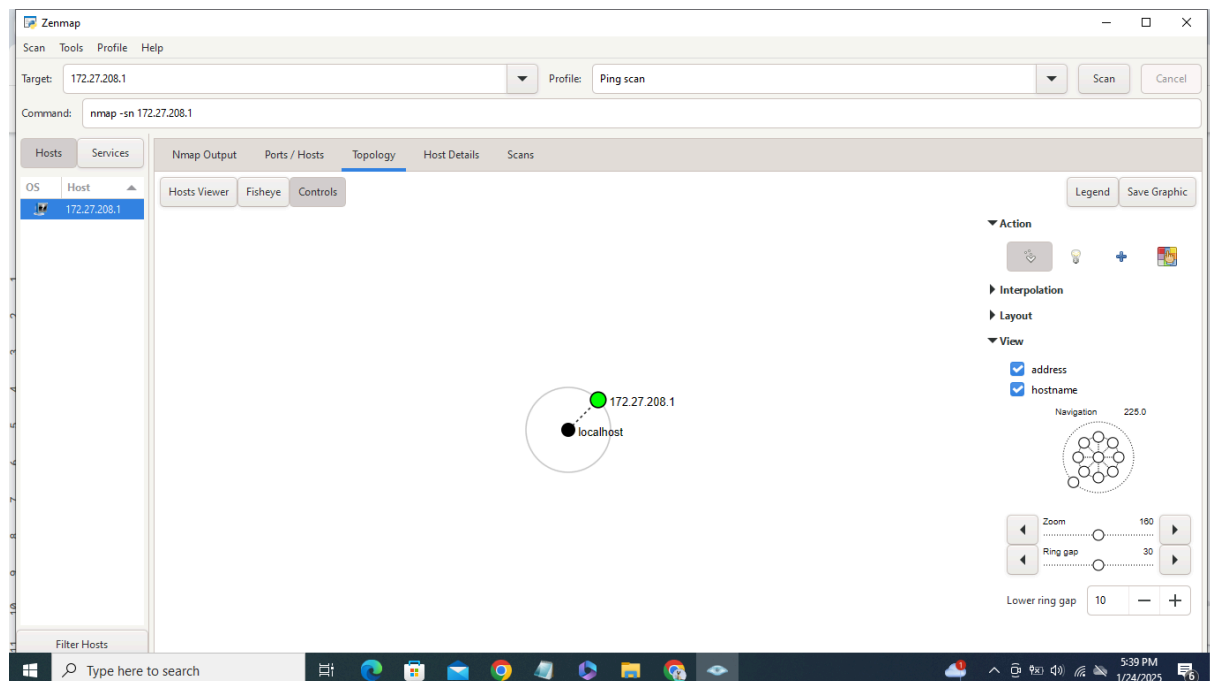
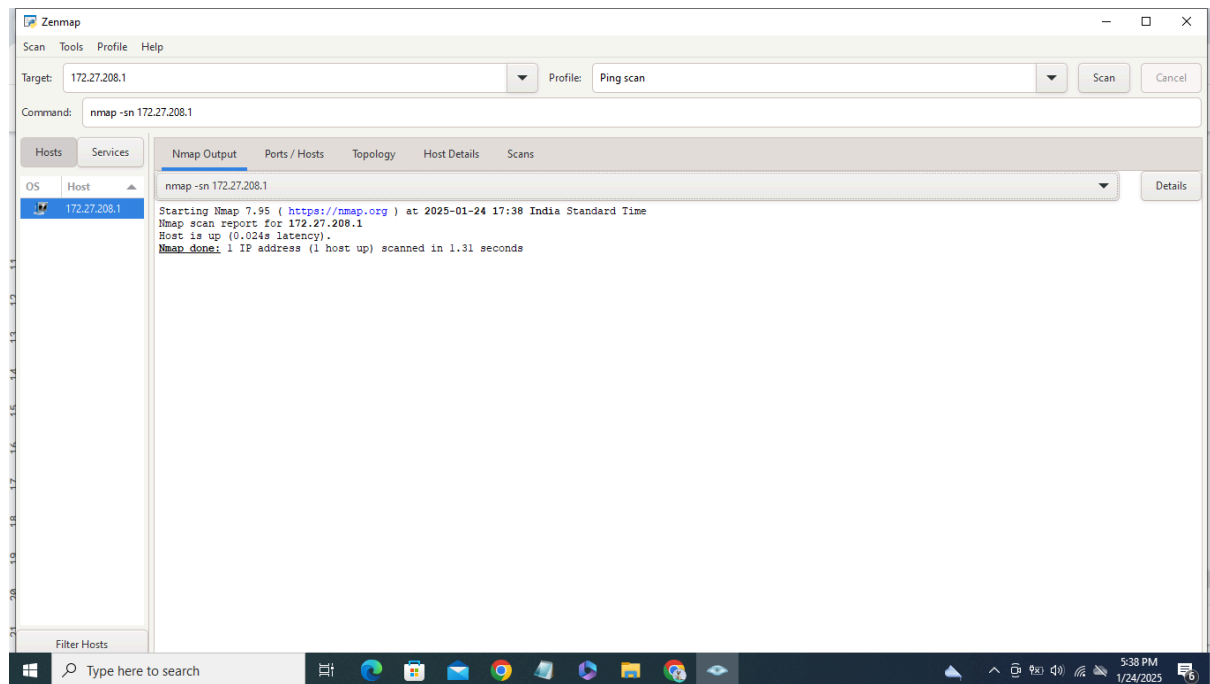
The Windows taskbar at the bottom shows the time as 8:33 PM on 1/23/2025.

The screenshot shows the Zenmap application window with the 'Topology' tab selected. The 'Hosts Viewer' is active, displaying a visual representation of the traceroute path. The path starts at 'localhost' and goes through several hops to reach 'google.com (142.250.193.238)'. The hops are labeled with their IP addresses and RTT values: 192.168.25.166, 255.0.0.0, 255.0.0.3, and dell1s18-in-fl4.1e100.net (142.250.193.238). The 'Fisheye on ring' slider is set to 1.00. The 'with interest factor' slider is set to 2.00, and the 'spread factor' is set to 0.50. The 'Navigation' button is visible in the top right corner.

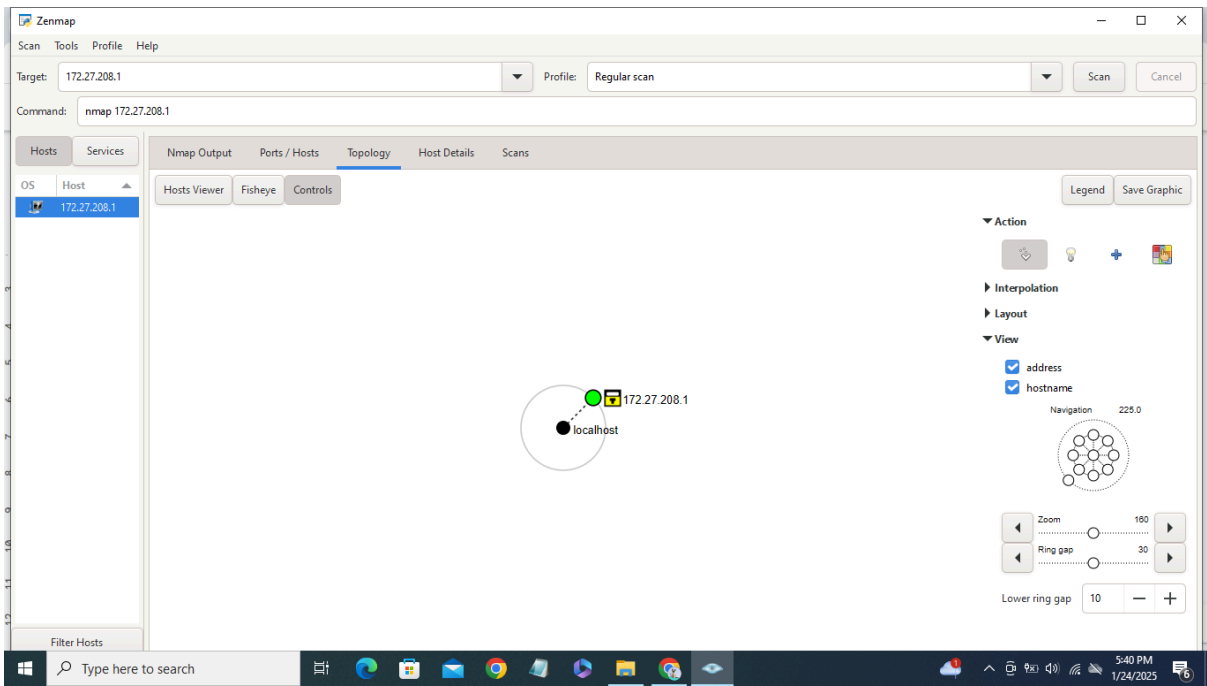
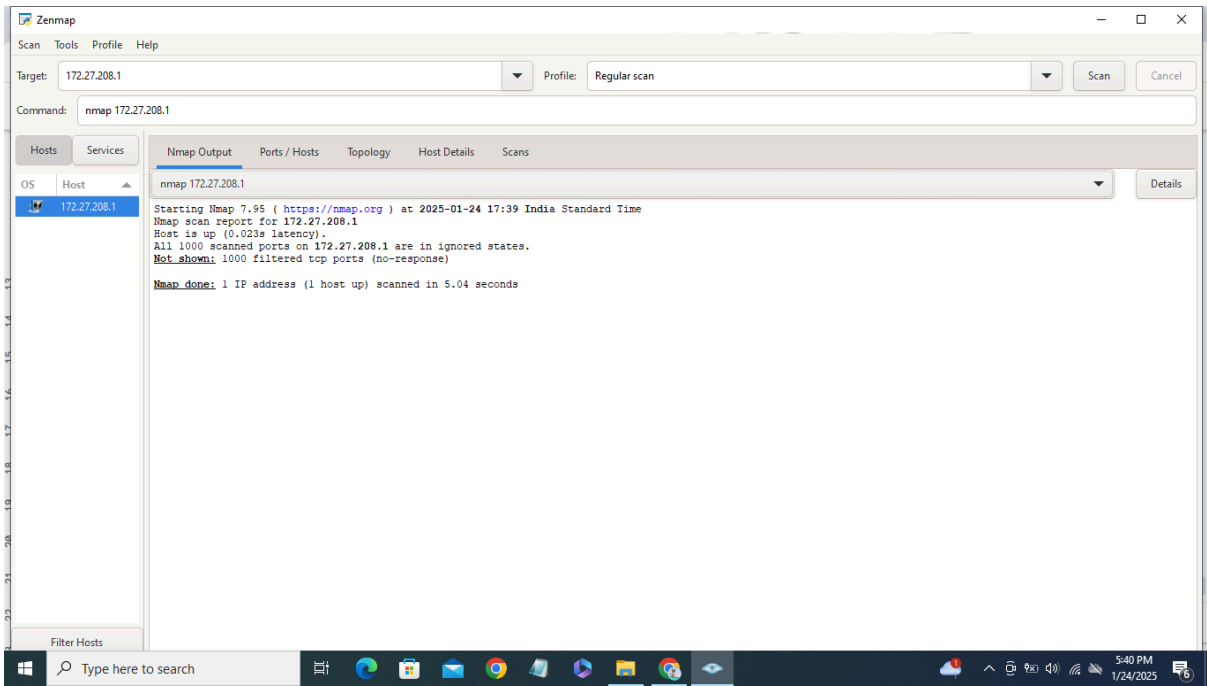
172.27.208.1 (Intense Scan,all TCP ports) (Lab PC IP) :



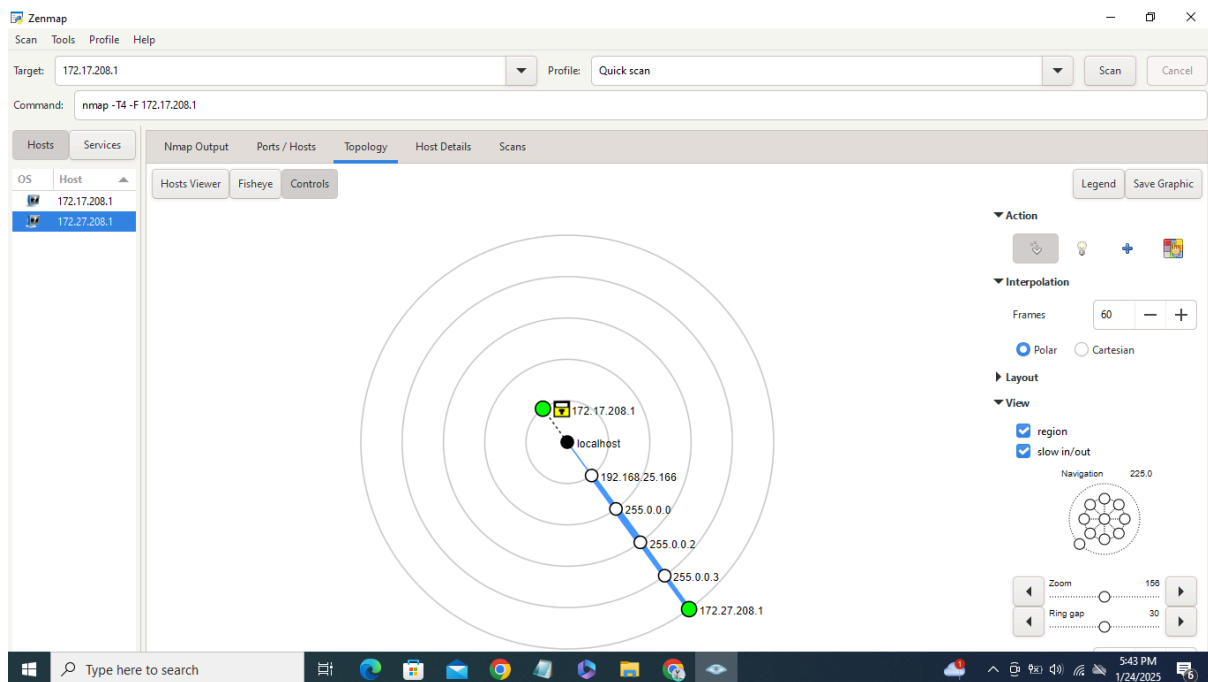
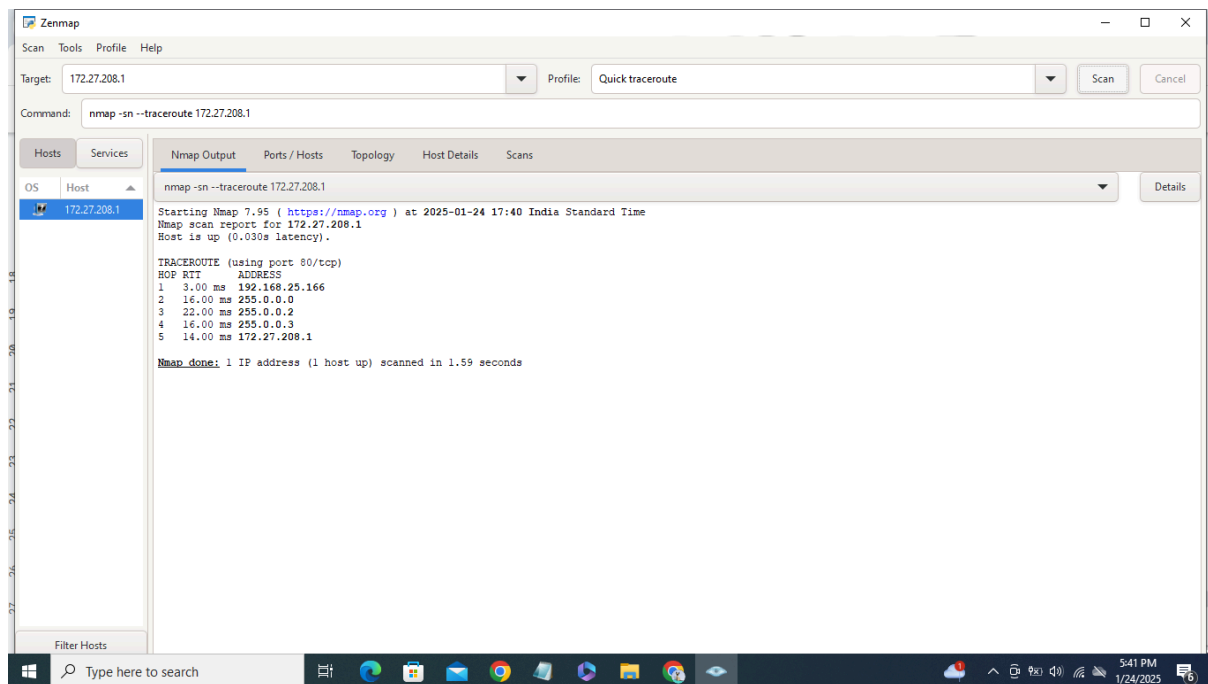
172.27.208.1 (Ping Scan)



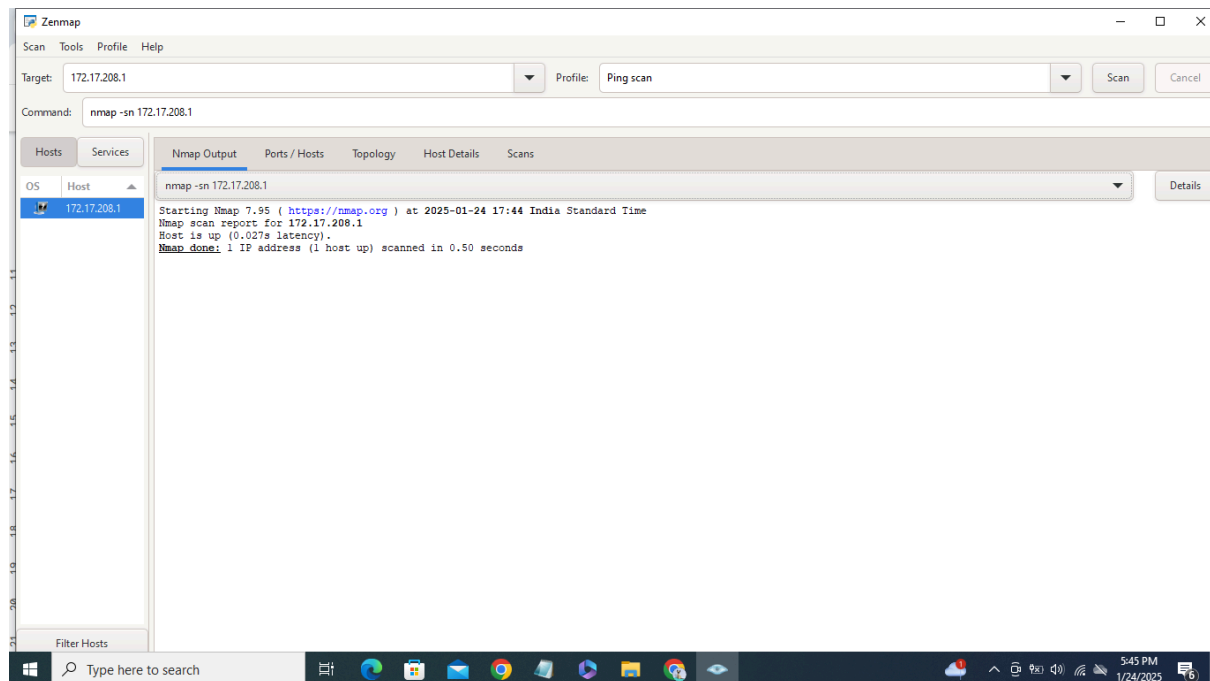
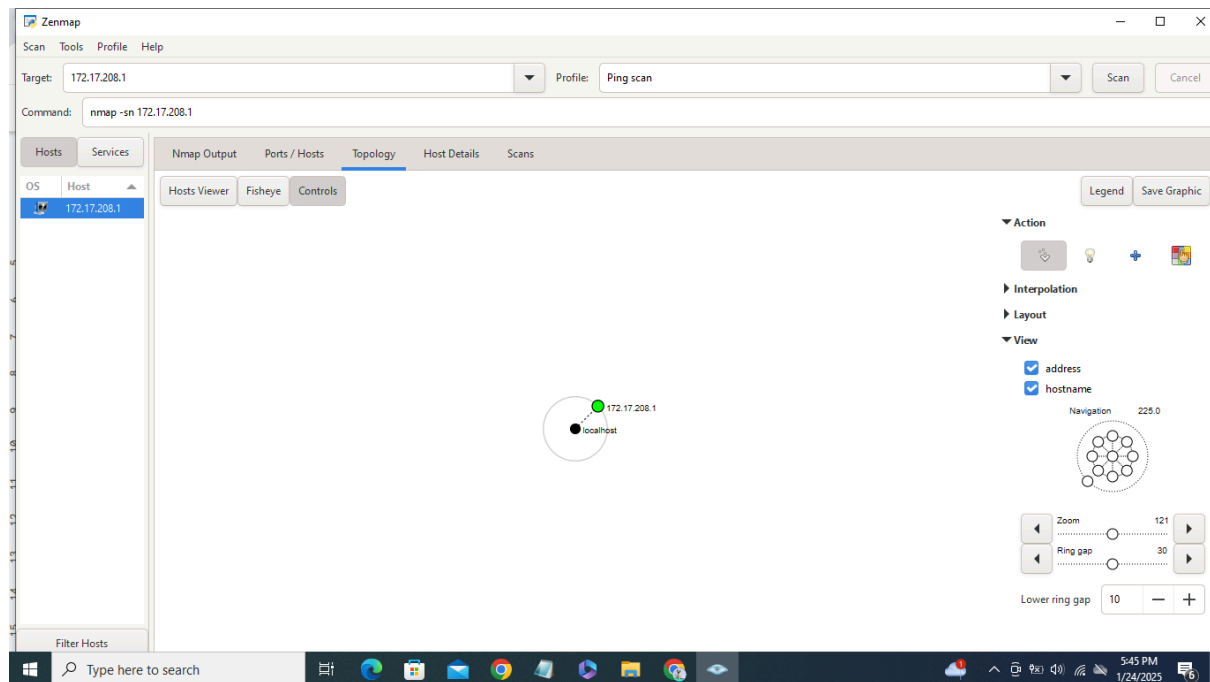
172.27.208.1 (Regular Scan)



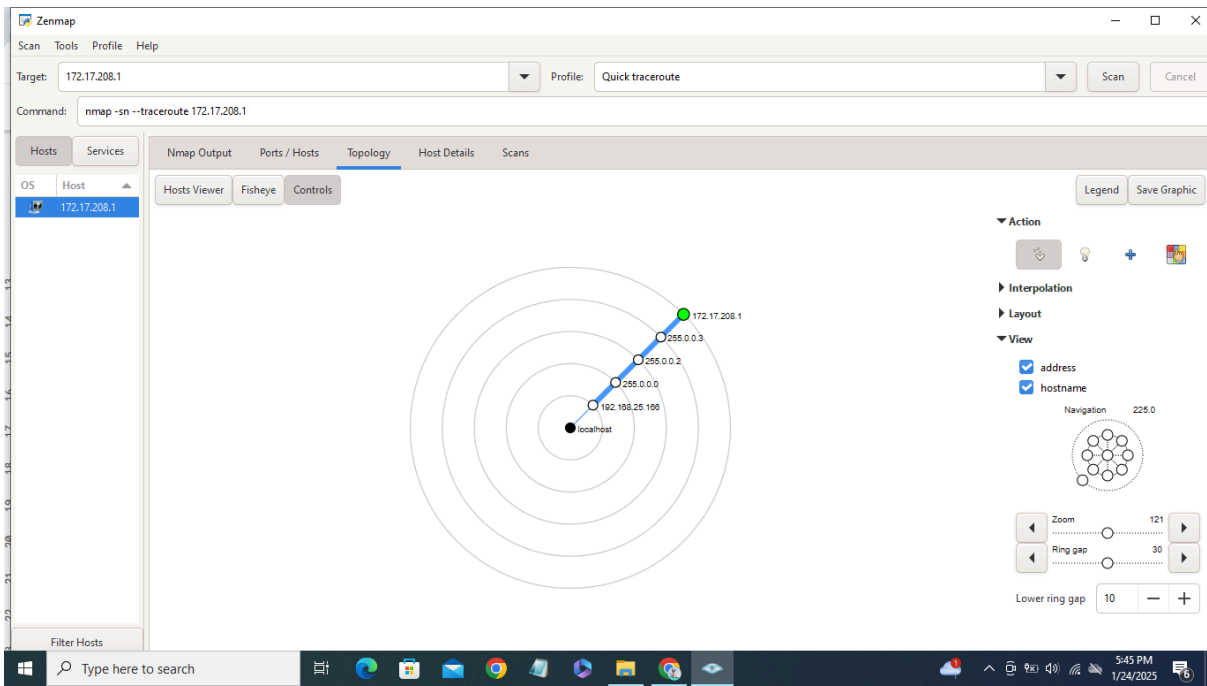
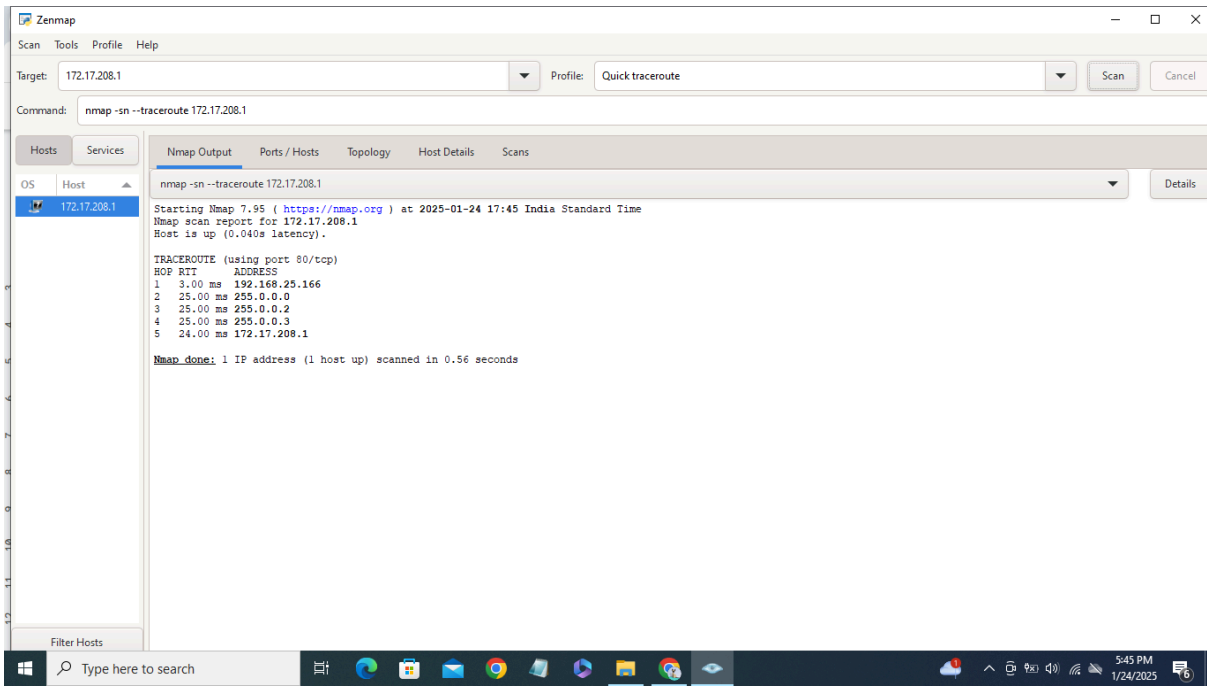
172.27.208.1 (Quick Traceroute Scan)



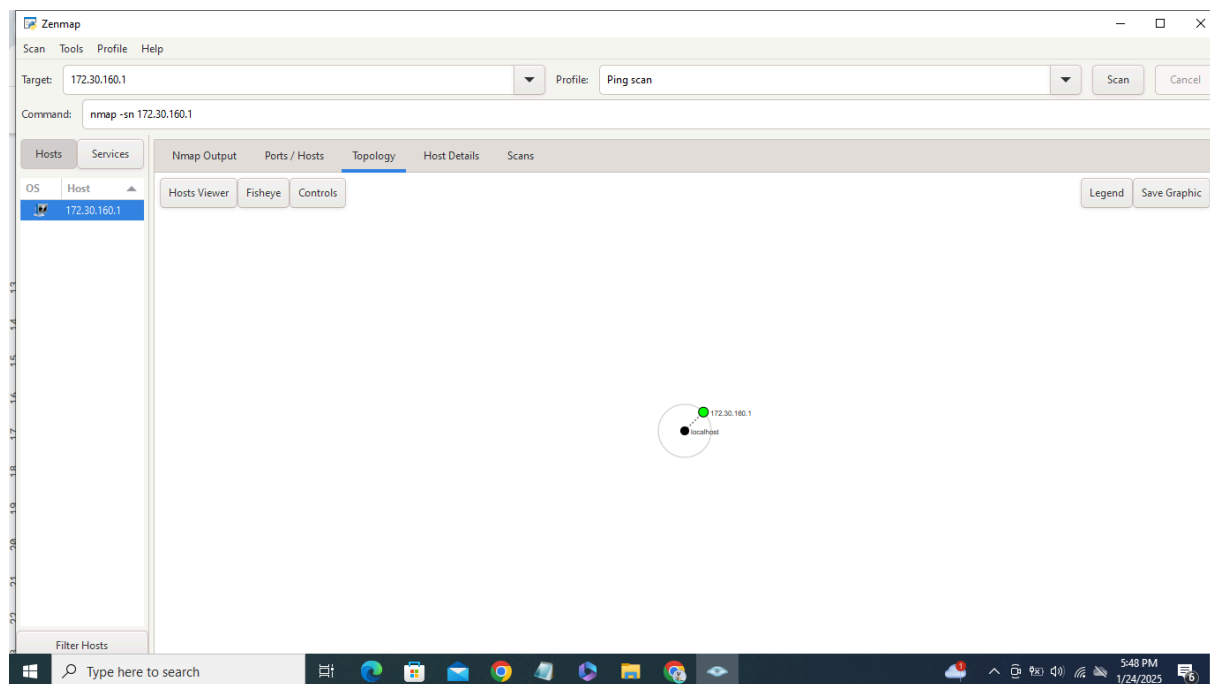
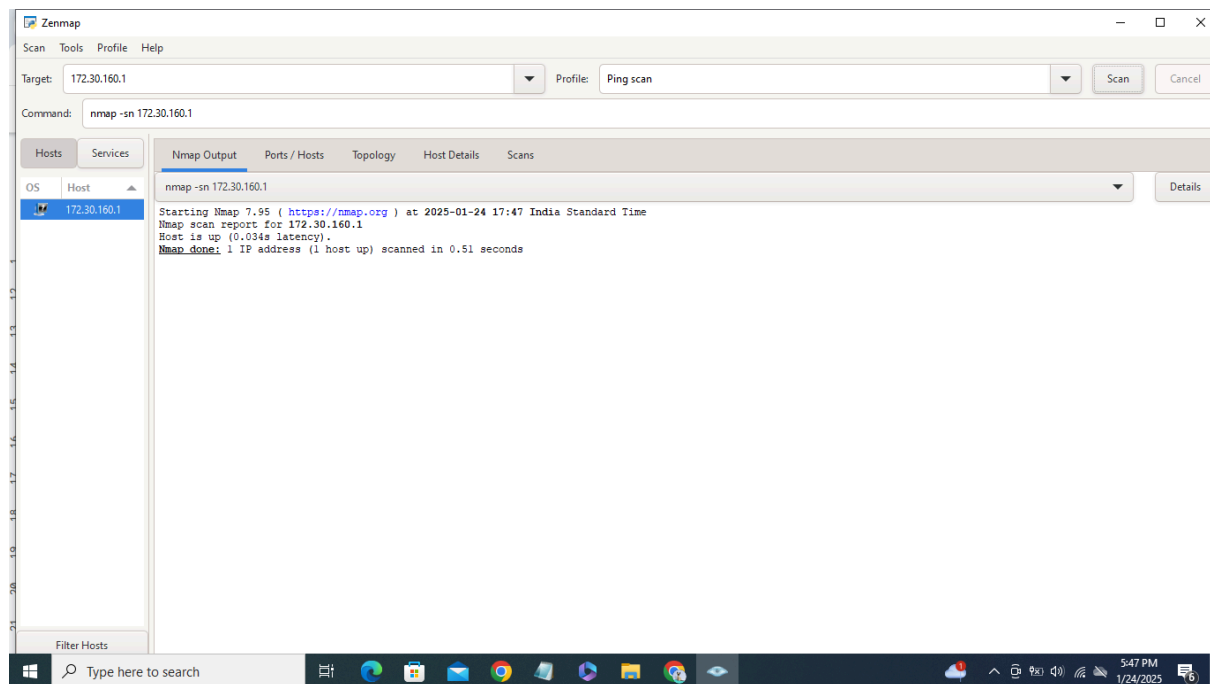
172.17.208.1 (Ping Scan) (Lab PC 2)



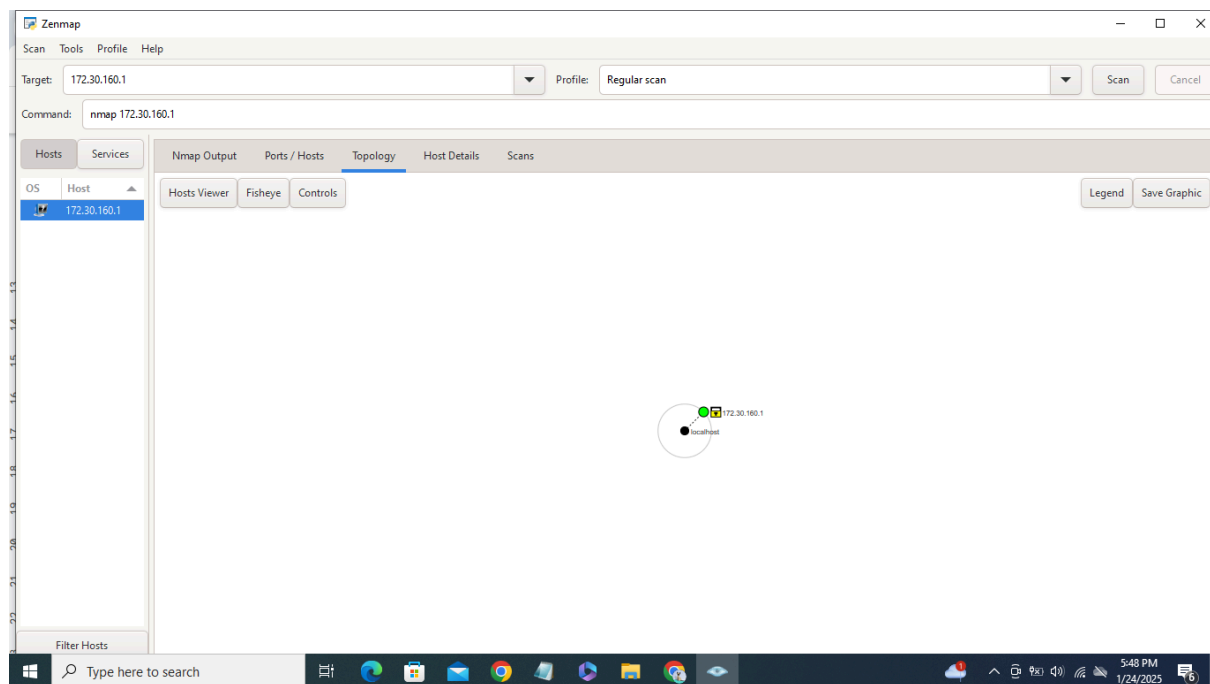
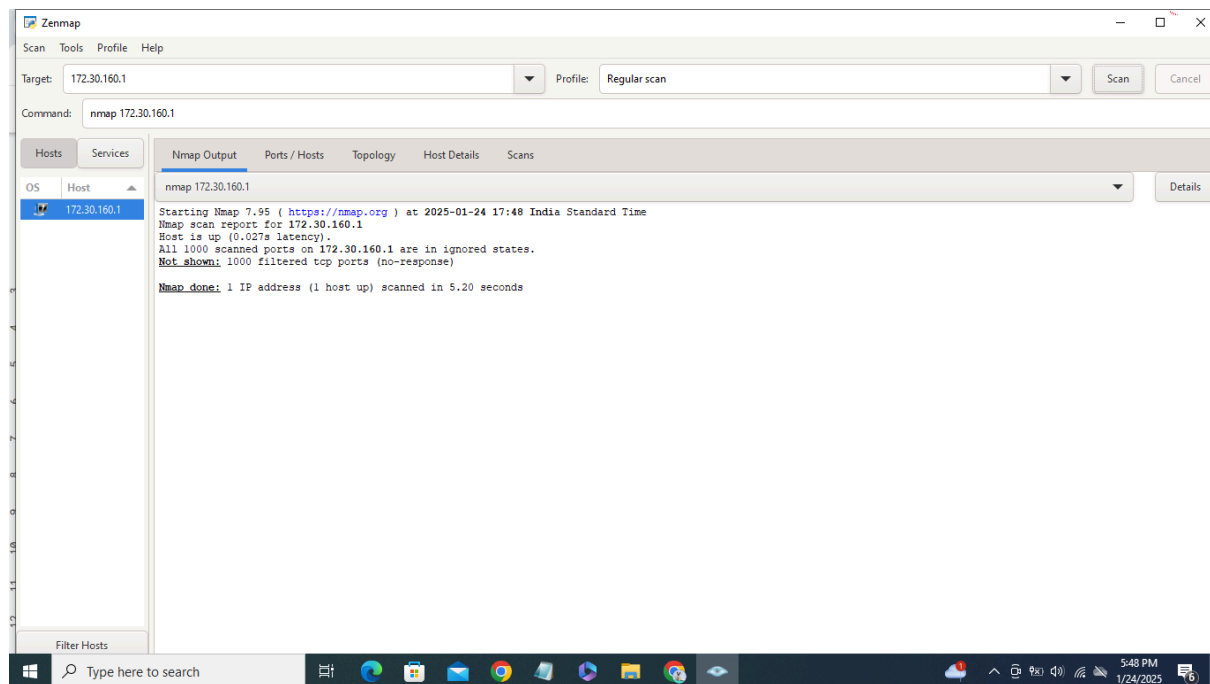
172.17.208.1 (Quick Traceroute Scan) (Lab PC 2)



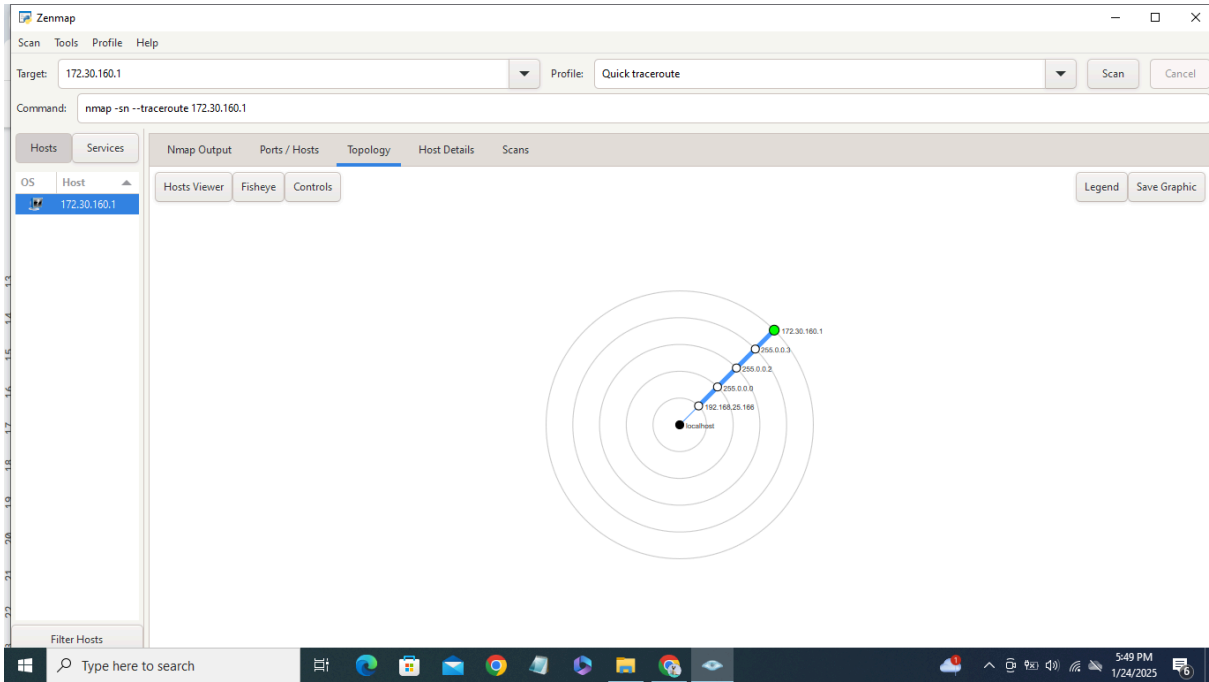
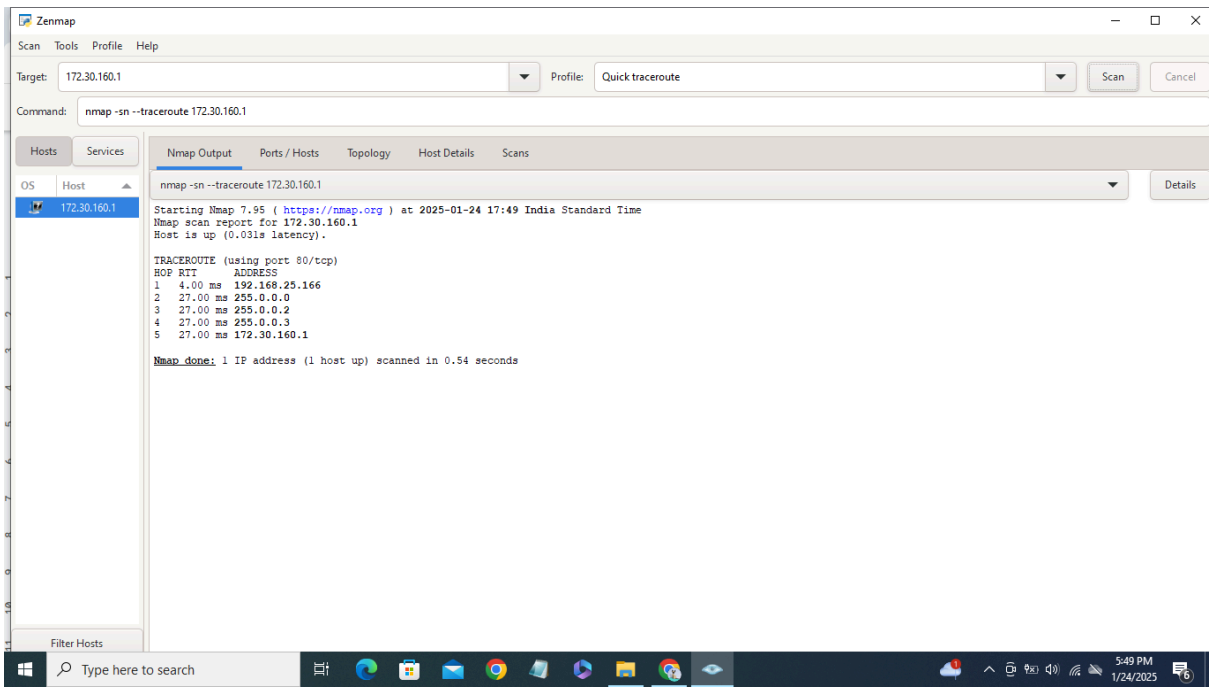
172.80.160.1 (Ping Scan) (Lab PC 3)



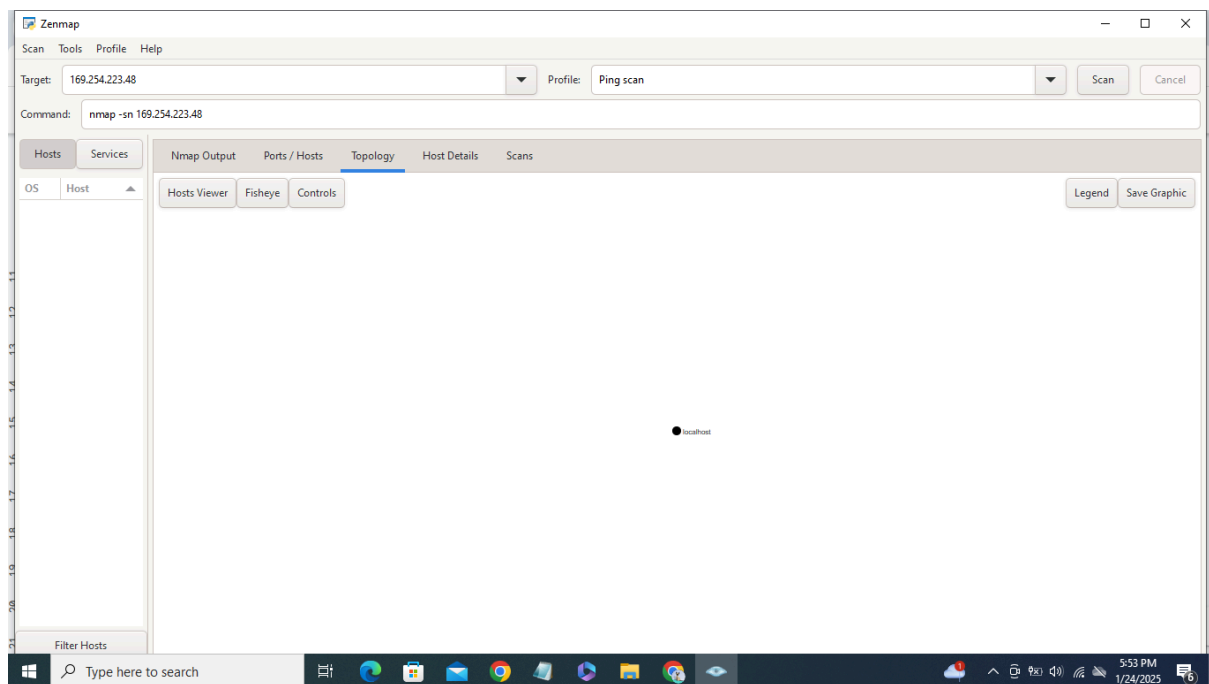
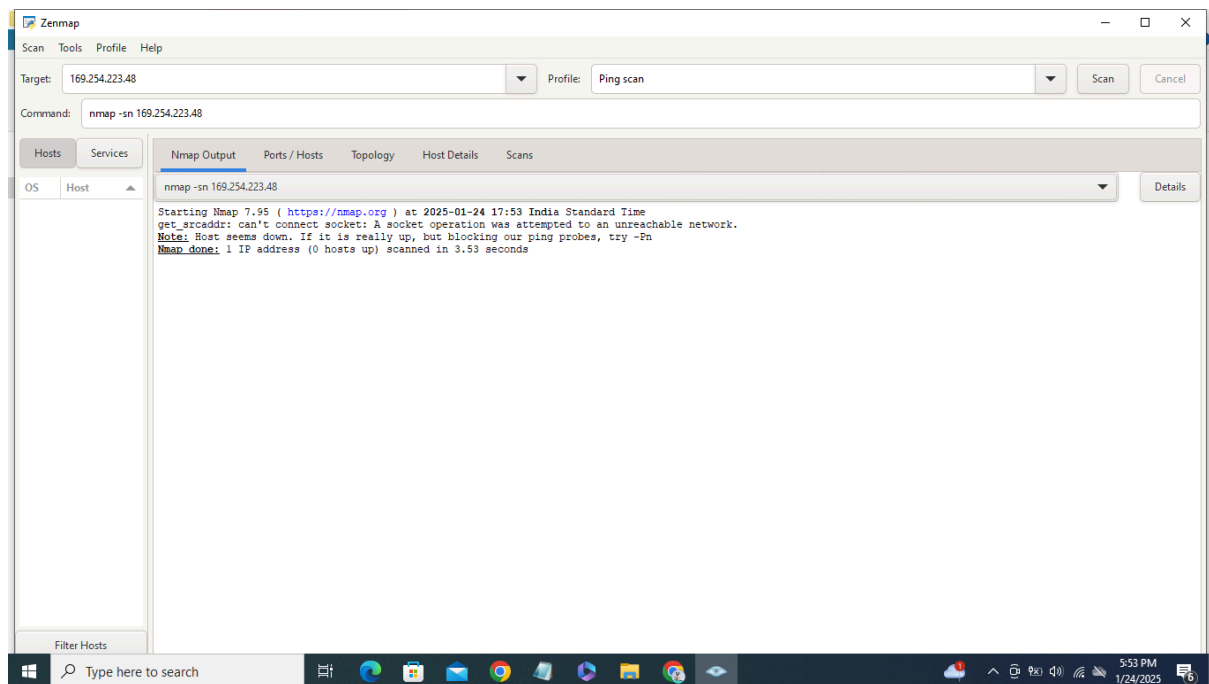
172.80.160.1 (Regular Scan) (Lab PC 3)



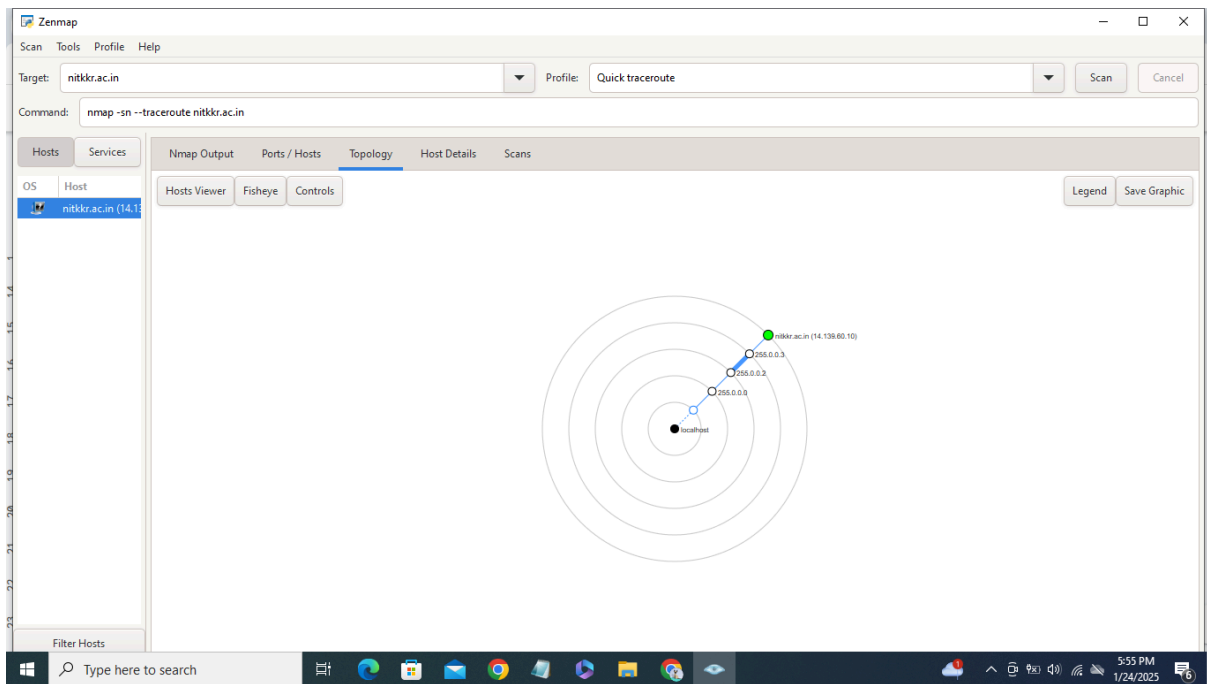
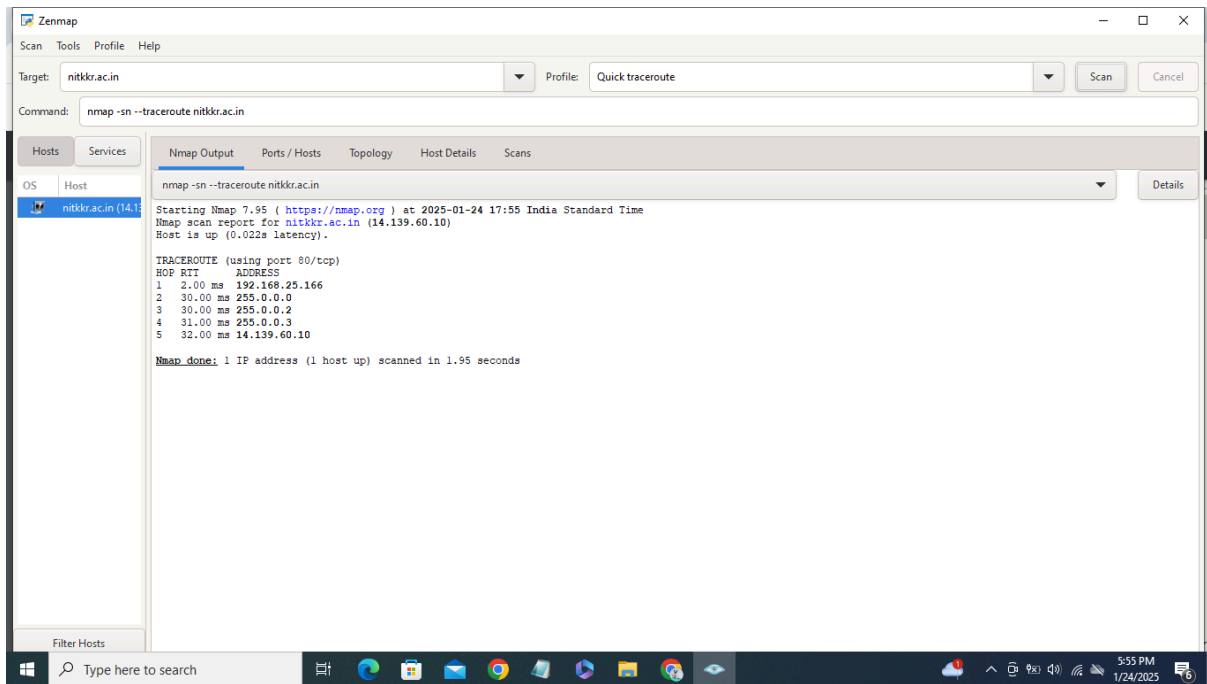
172.80.160.1 (Quick Traceroute Scan) (Lab PC 3)



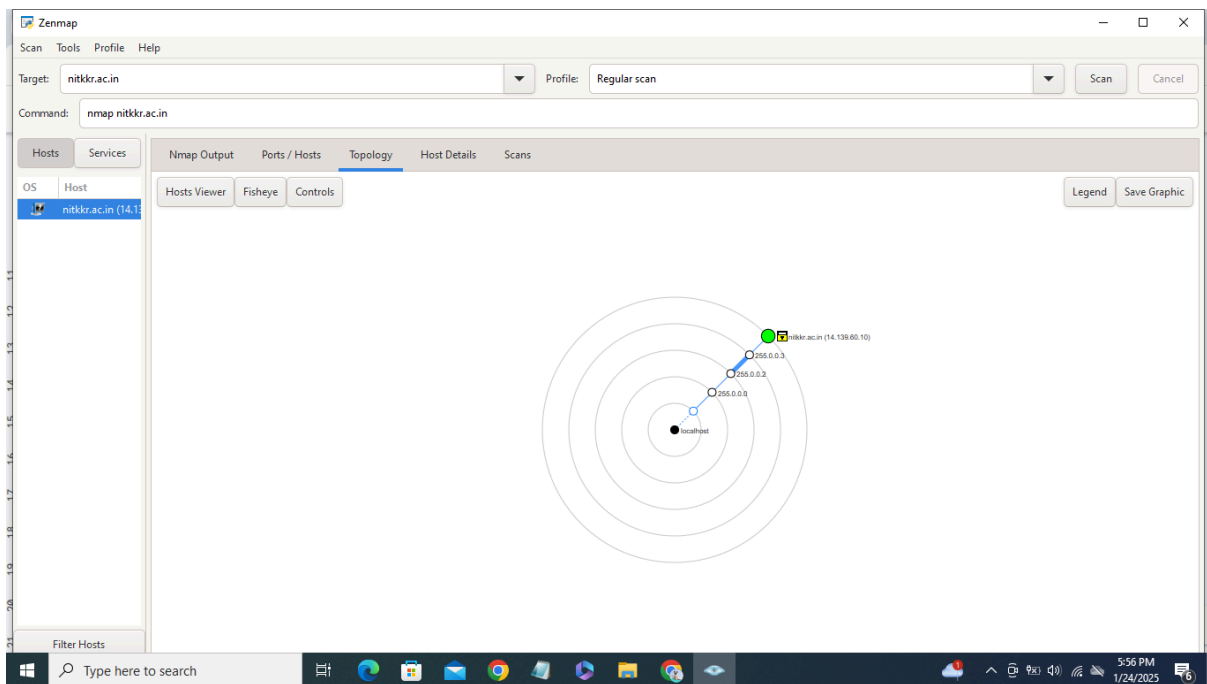
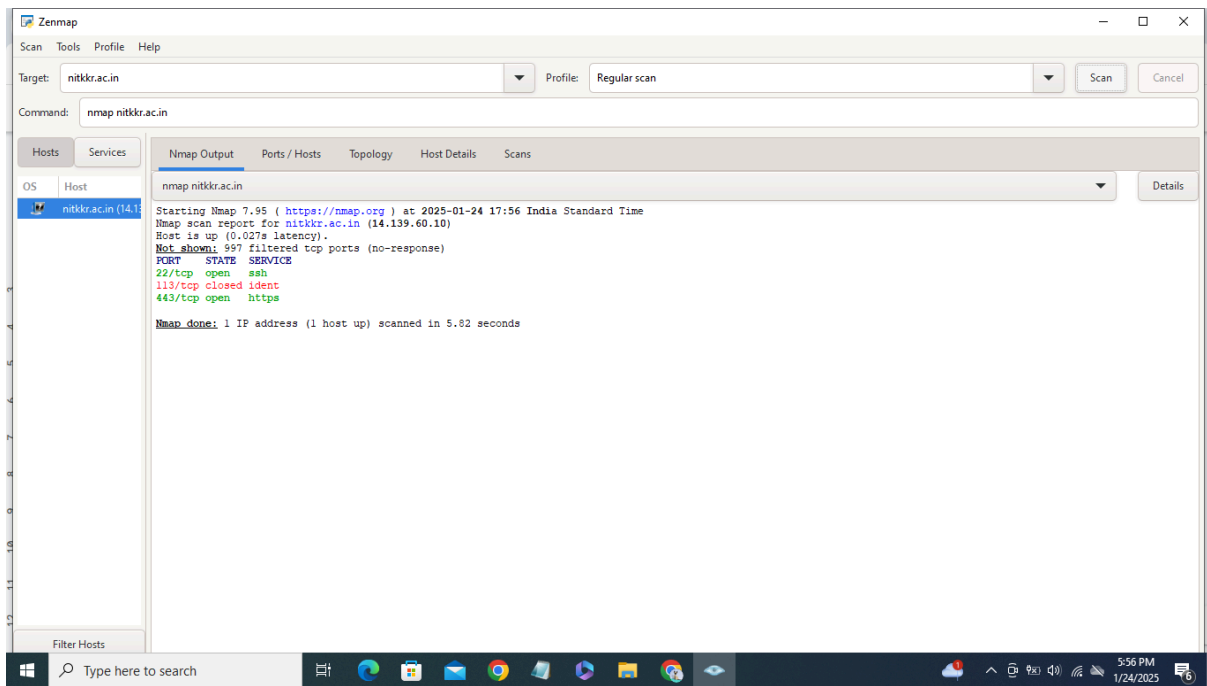
169.254.223.48 (Ping Scan) (Next Lab PC)



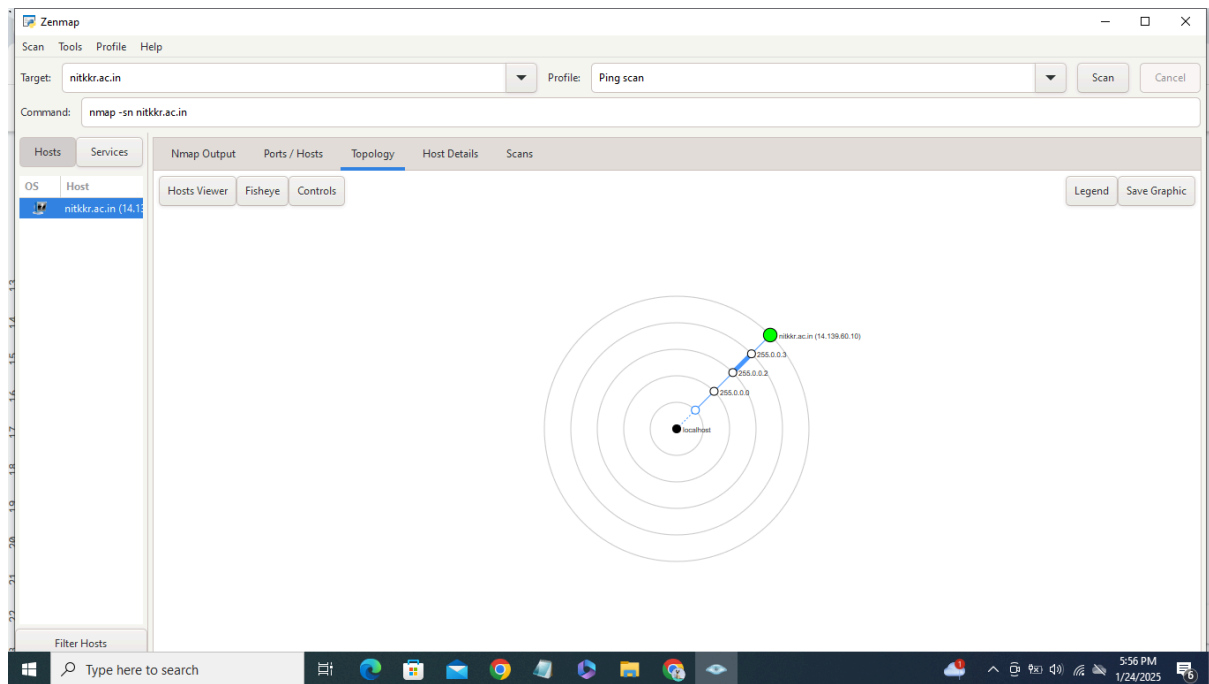
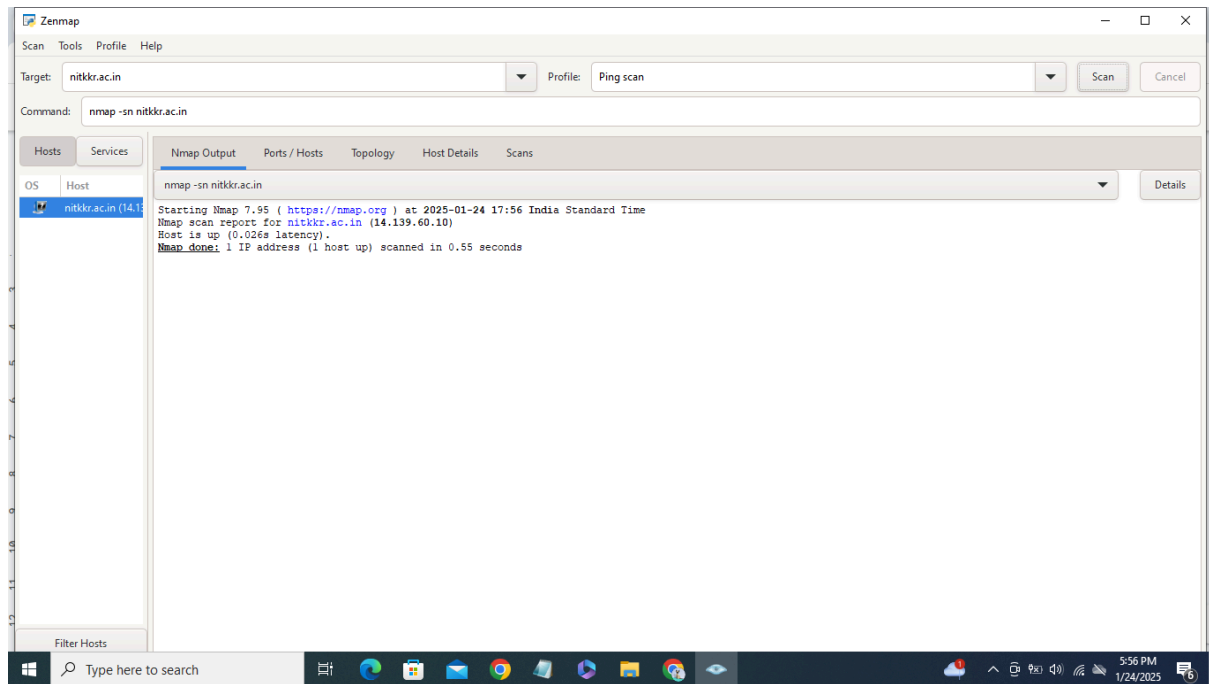
nitkkr.ac.in (Quick Traceroute Scan)



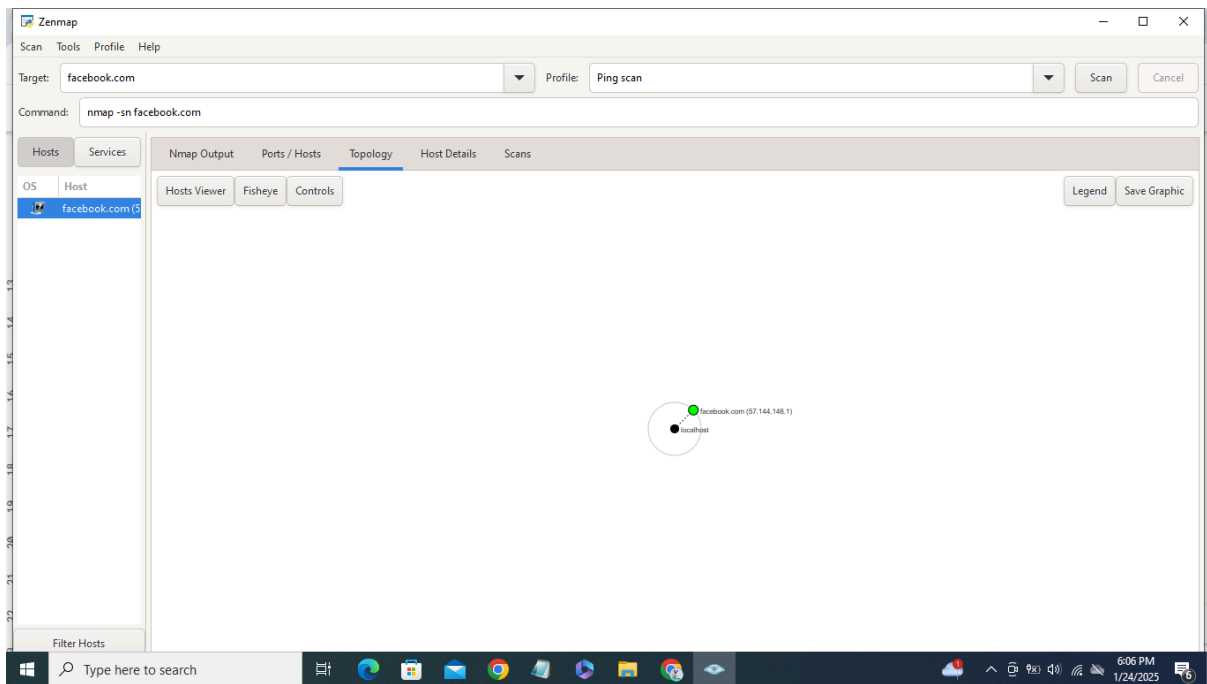
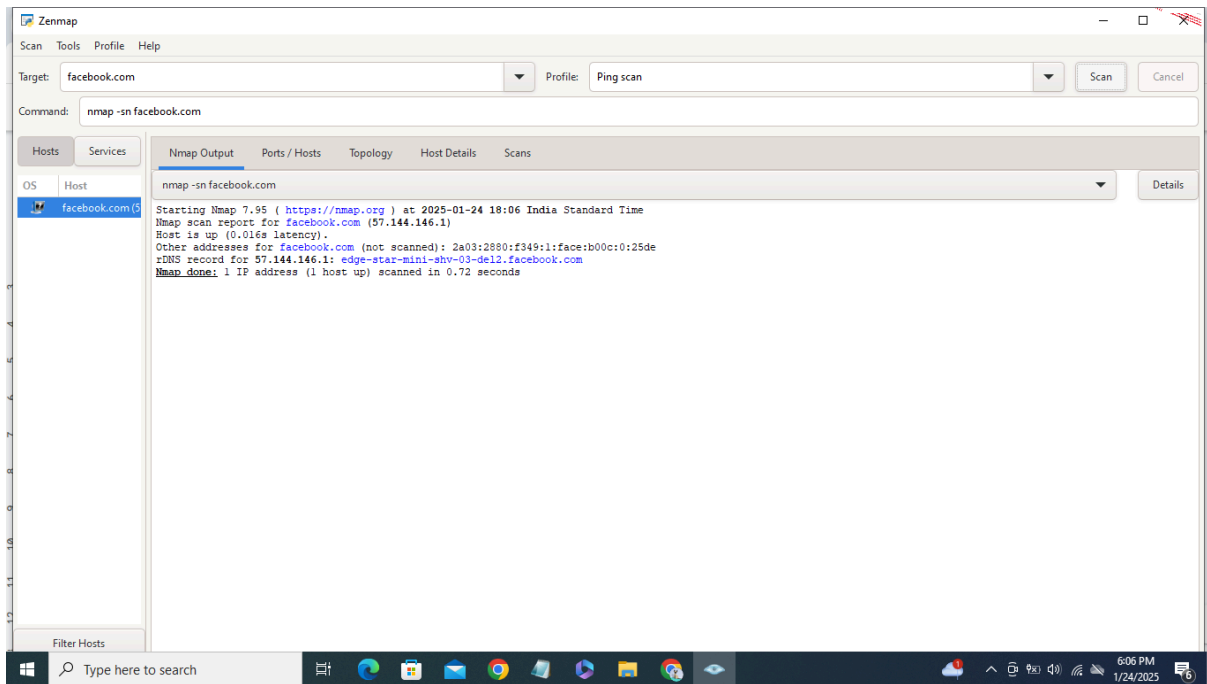
nitkkr.ac.in (Regular Scan)



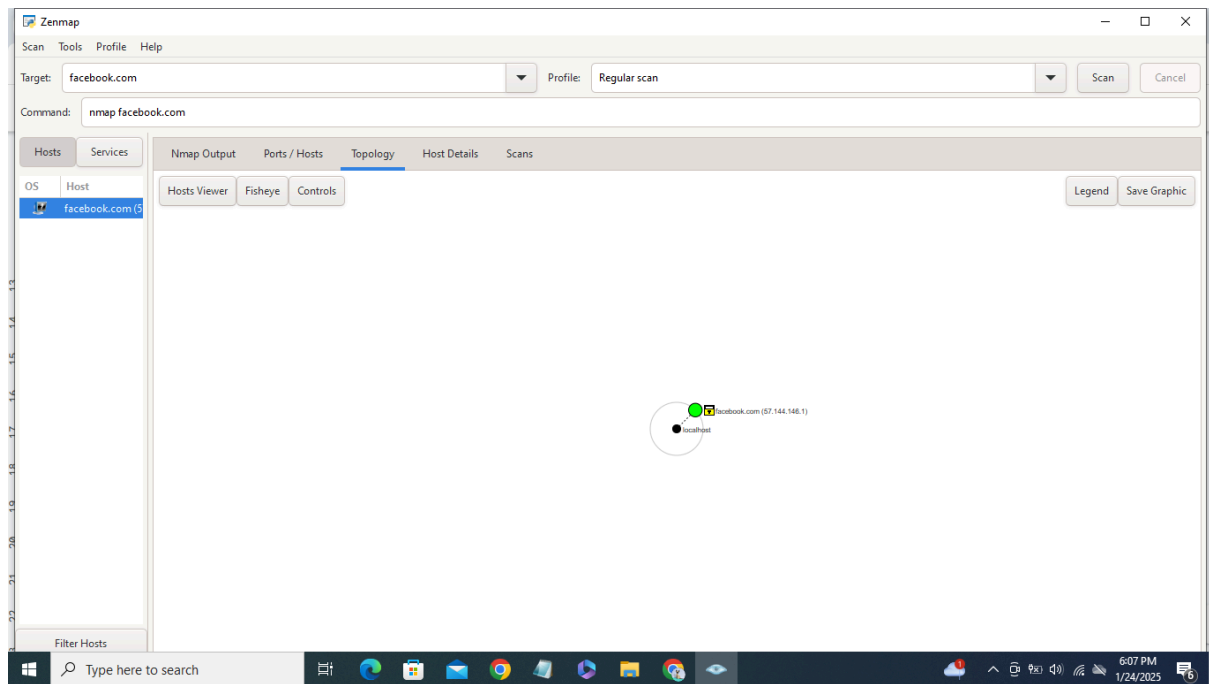
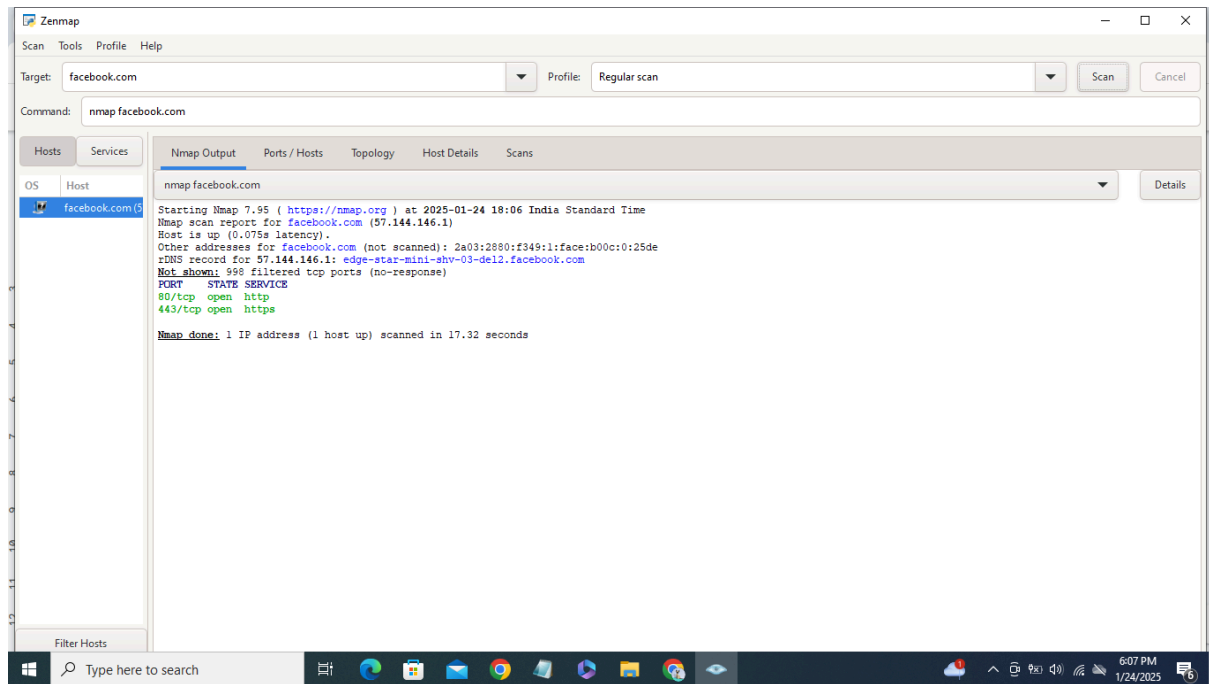
nitkkr.ac.in (Ping Scan)



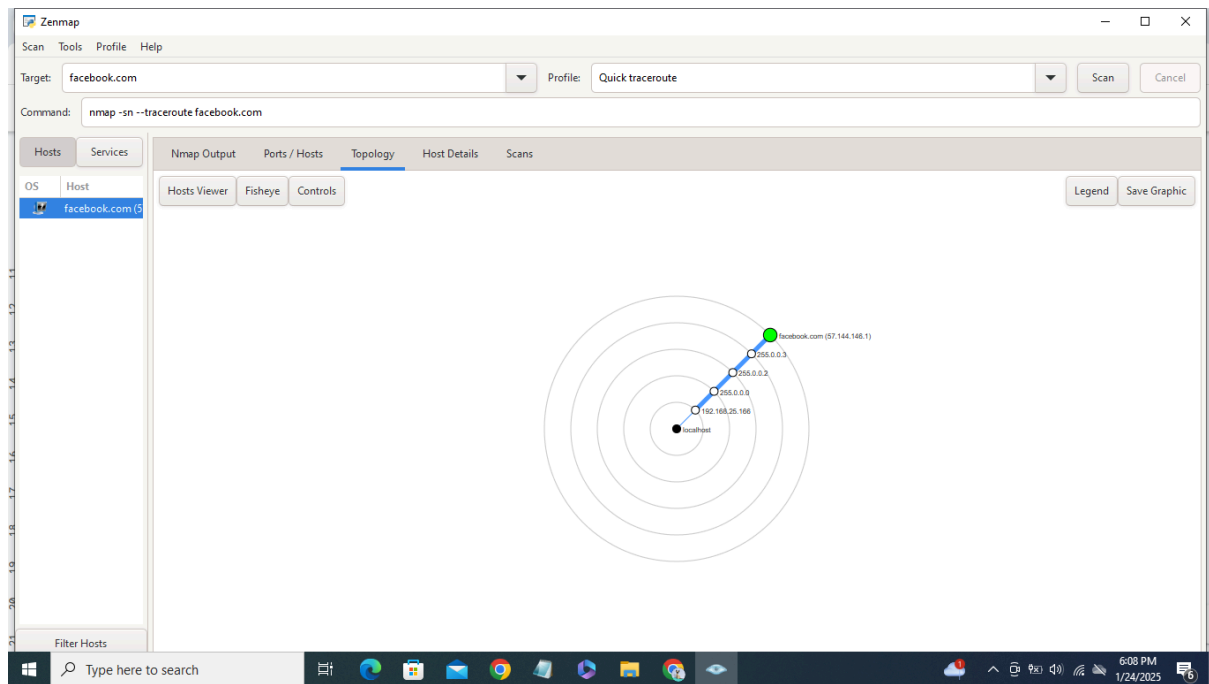
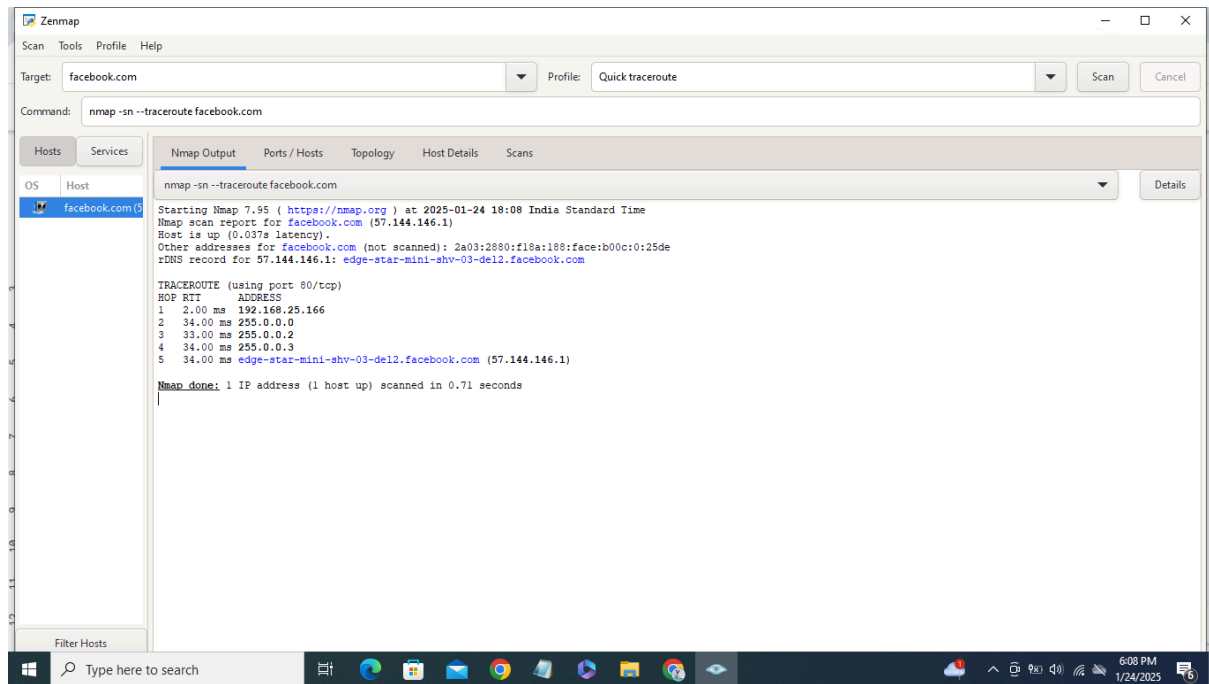
facebook.com (Ping Scan)



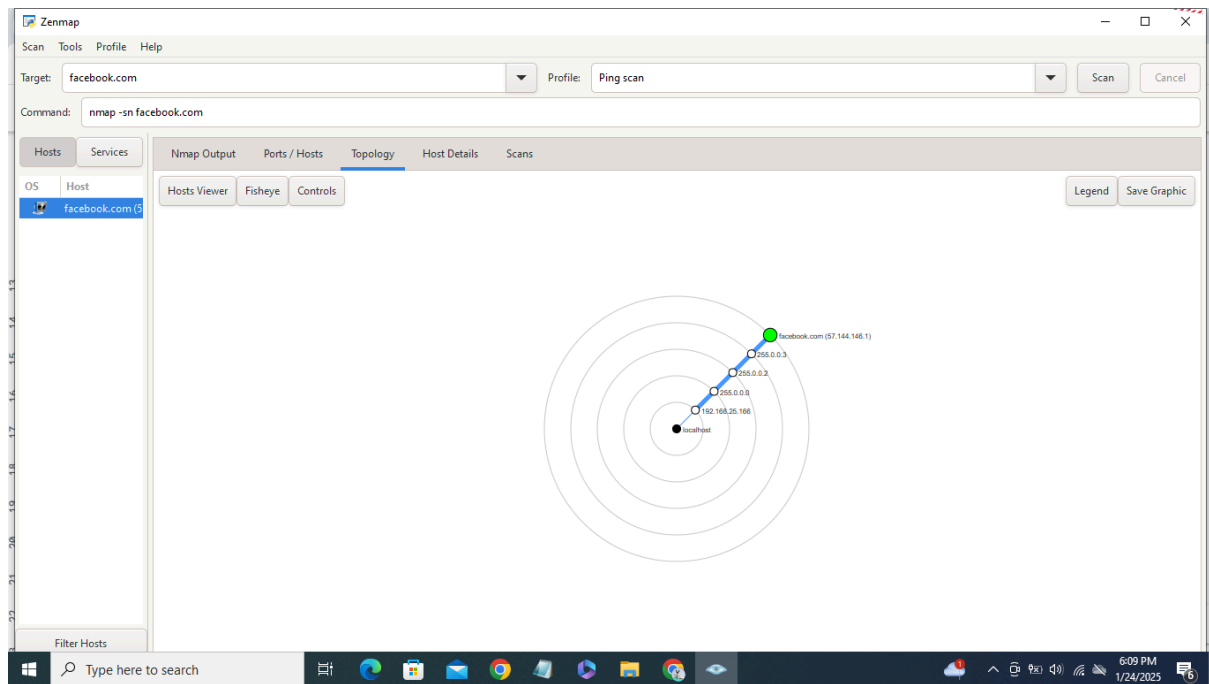
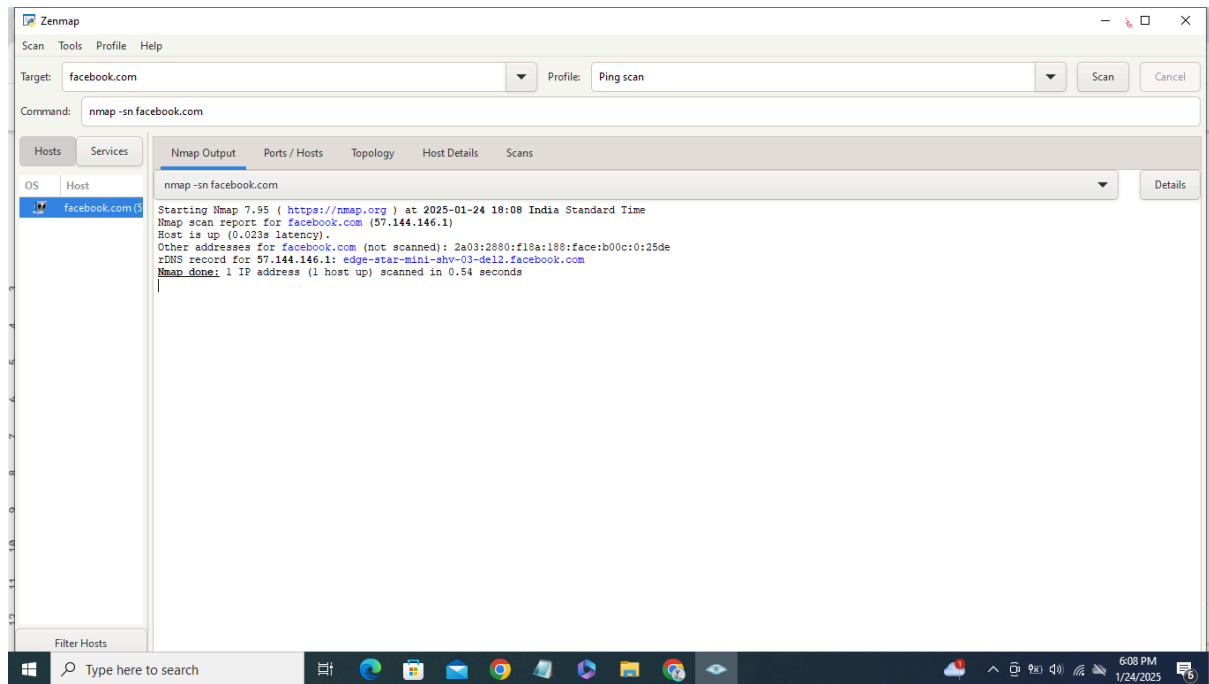
facebook.com (Regular Scan)



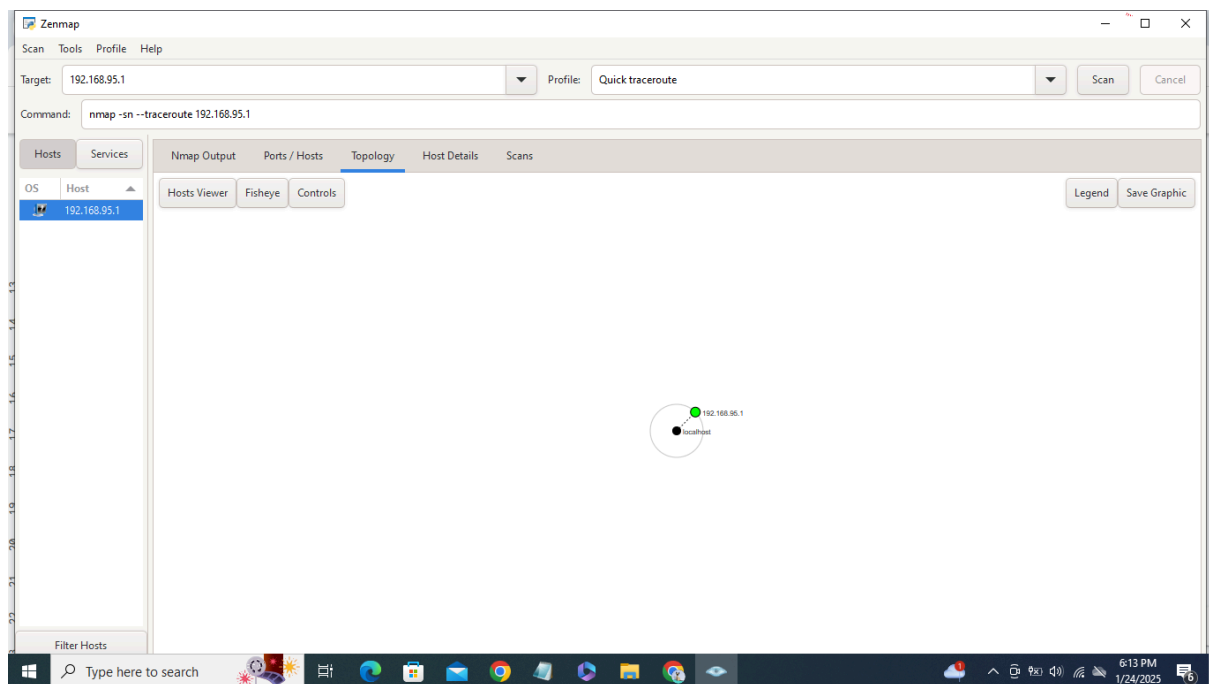
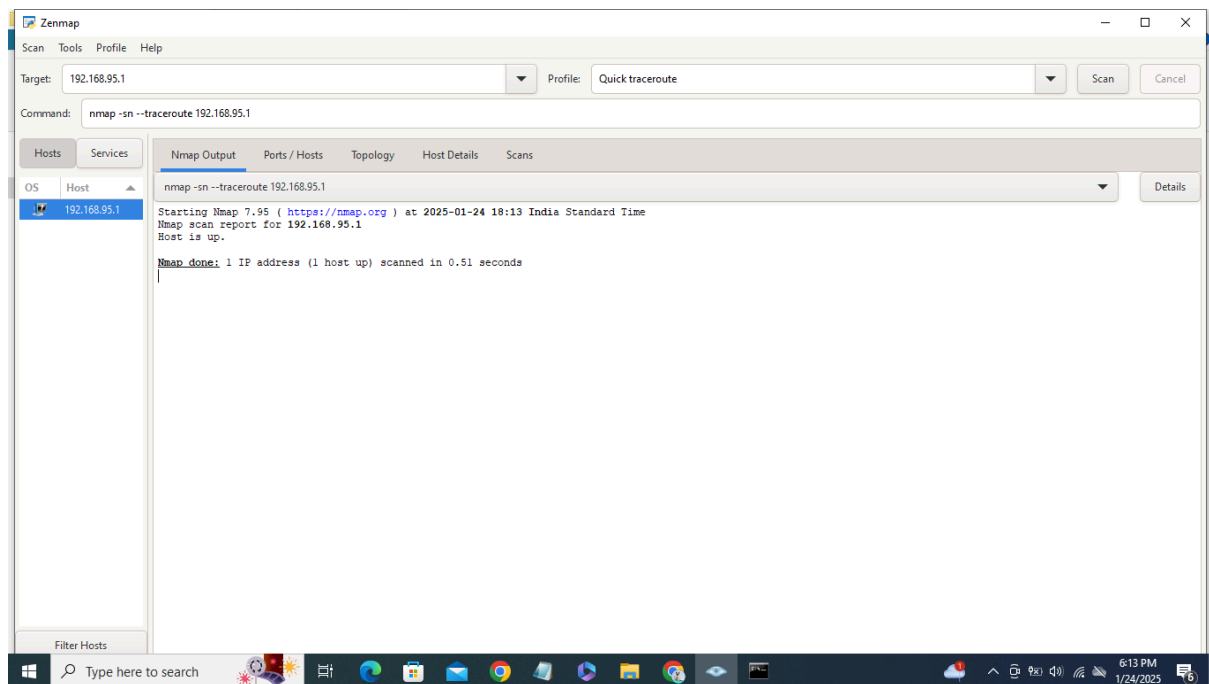
facebook.com (Quick Traceroute Scan)



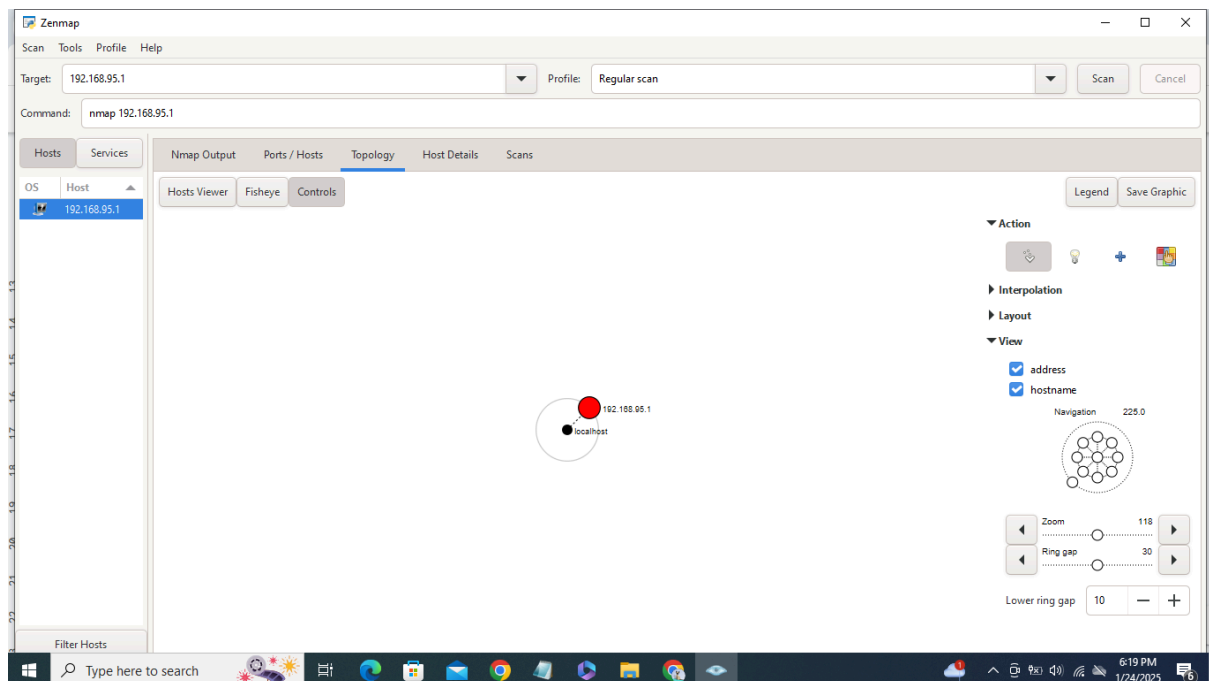
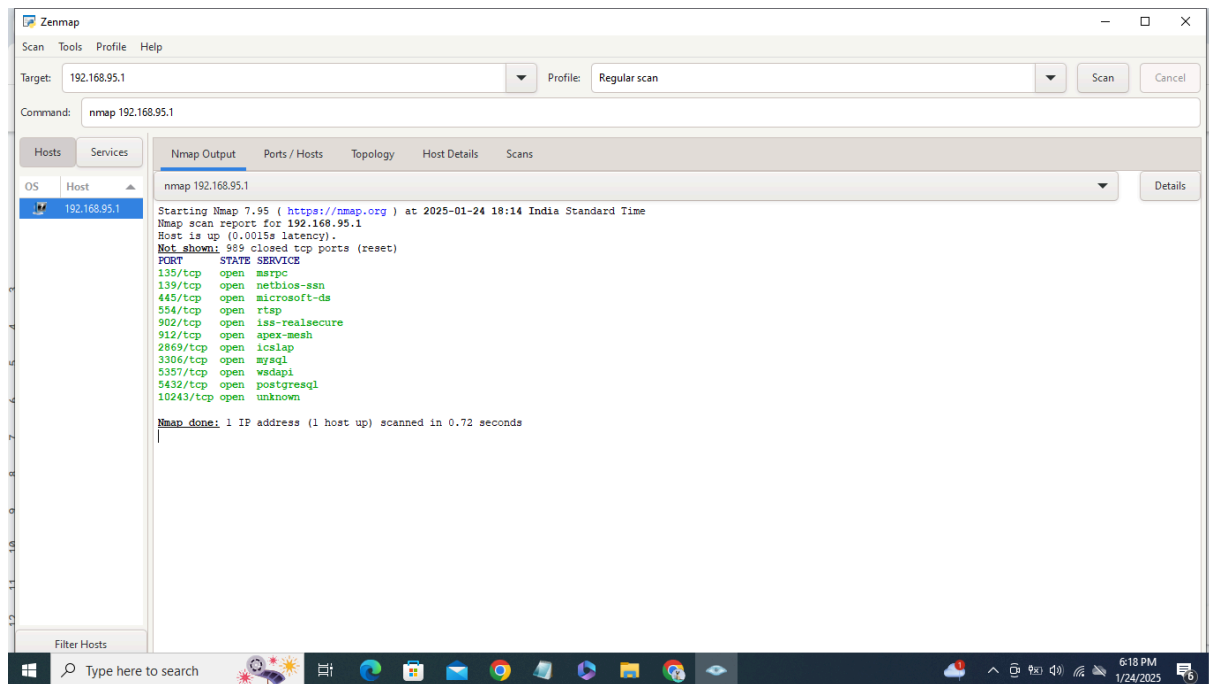
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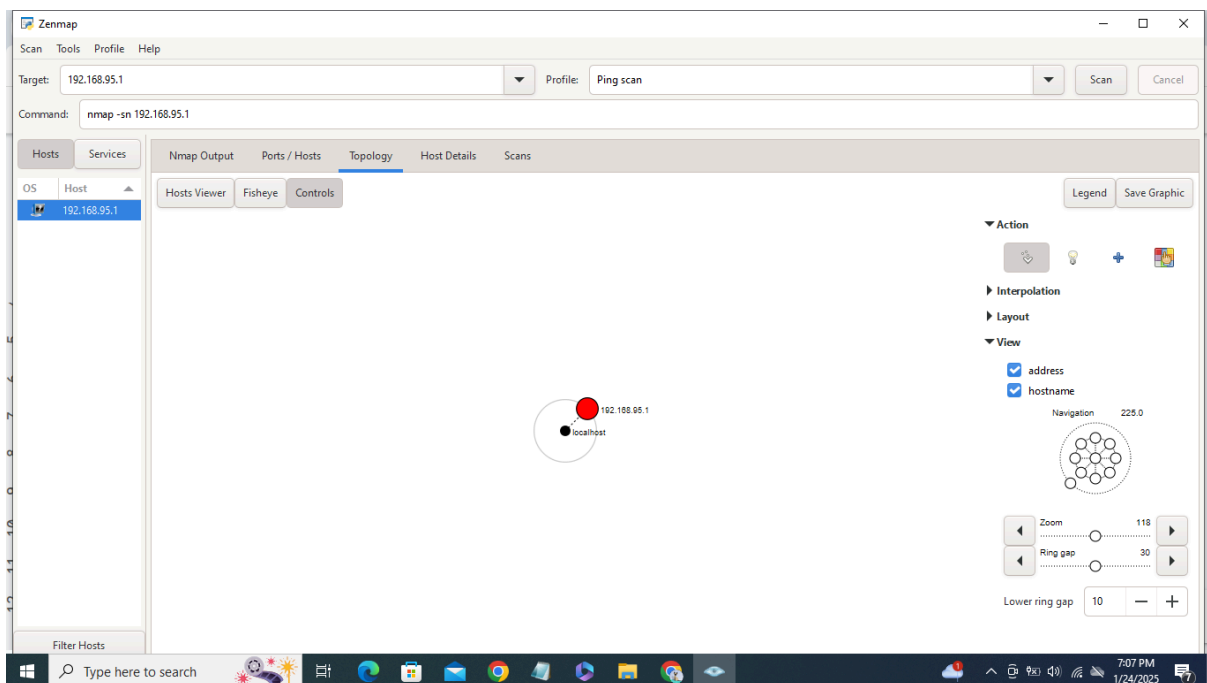
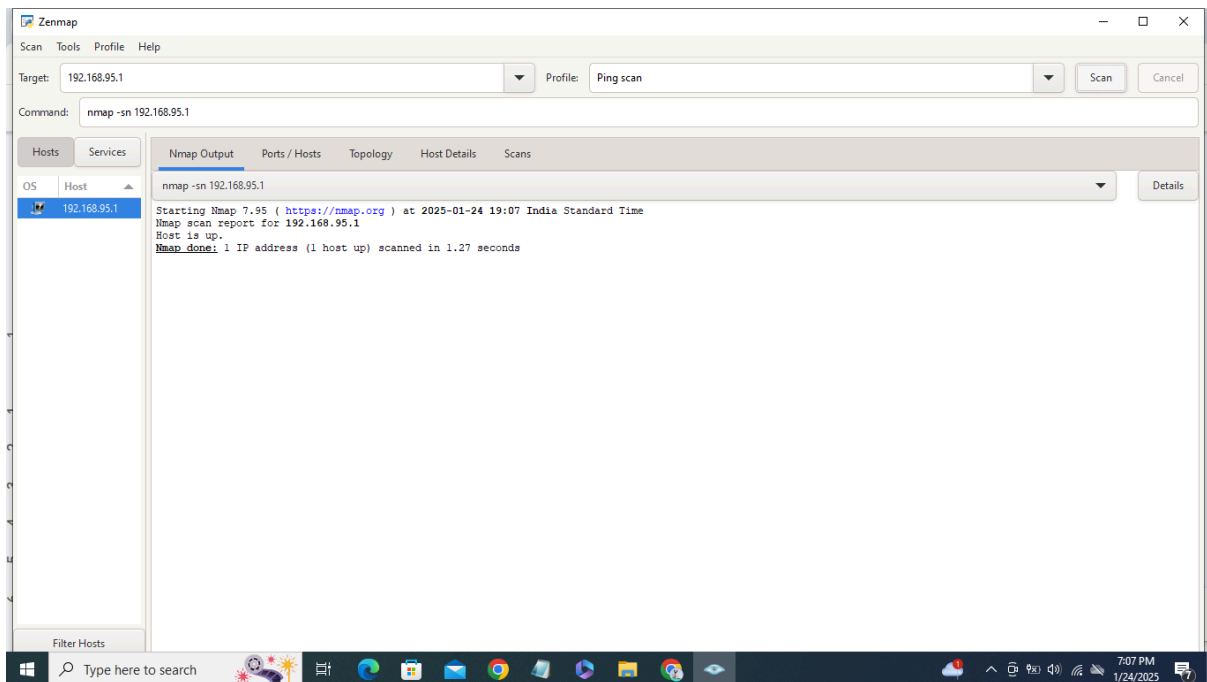
192.168.95.1 (Quick Traceroute Scan) (Outside of NIT PC)



192.168.95.1 (Regular Scan) (Outside of NIT PC)



192.168.95.1 (Quick Traceroute Scan) (Outside of NIT PC)



192.168.95.1 (Specific Port Scan) (Outside of NIT PC)

